

December 2018

ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 70***Answer to Question No.1 is compulsory and to be answered first.****This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.****Answer any five questions from the rest.**

1. A. Choose the most appropriate answer from the following (any ten): 1×10
- i) Which of the following material has maximum ductility? - (a) Mild steel, (b) Copper, (c) Nickel, (d) Aluminium.
 - ii) The hardness is the property of a material due to which it: - (a) can be drawn into wires, (b) breaks with little permanent distortion, (c) can cut another metal, (d) can be rolled or hammered into thin sheets.
 - iii) The percentage of carbon in cast iron varies from: - (a) 0.1 to 0.5, (b) 0.5 to 1, (c) 1 to 1.7, (d) 1.7 to 4.5.
 - iv) Rapid quenching of austenite in the heat treatment of steel produces a super saturated and distorted crystal structure which is known as: - (a) cementite, (b) pearlite, (c) martensite, (d) ferrite.
 - v) Which among the following is a para-magnetic material? - (a) Iron, (b) Cobalt, (c) Nickel, (d) Aluminium.
 - vi) Bronze is a alloy of: - (a) copper & zinc, (b) copper, tin & zinc, (c) copper & tin, (d) none of these.
 - vii) Dye penetrant method is generally used to locate- (a) core defects, (b) surface defects, (c) superficial defects, (d) temporary defects.
 - viii) Silicon when added to copper improves: - (a) machinability, (b) hardness, (c) hardness and strength, (d) strength and ductility.
 - ix) The lower critical point for all steels is: - (a) 600°C, (b) 700°C, (c) 723°C, (d) 913°C.
 - x) The heat treatment process used for softening hardened steel is: - (a) carburizing, (b) normalizing, (c) annealing, (d) tempering.
 - xi) The hardness of steel increases if it contains: - (a) pearlite, (b) ferrite, (c) cementite, (d) martensite.
 - xii) Non-destructive testing is used to determine: - (a) location of defects, (b) chemical composition, (c) corrosion of metal, (d) All of these.
 - xiii) Which of the following tool is manufactured by powder metallurgy?: - (a) High speed steel, (b) Sintered carbides, (c) High carbon steel, (d) Low carbon steel.

B. State whether the following statements are true or false (any ten):

1×10

- i) Carbon is present in the form of graphite in gray cast iron.
- ii) Nickel increases strength and toughness in alloy steel.
- iii) Compression test is a type of non-destructive test.
- iv) Thermal conductivity of non metal is inversely proportional to temperature.
- v) Stiffness is the ability of a material to resist deformation under stress.
- vi) Pitting is a localized form of corrosion.
- vii) An eutectoid steel consists of pearlite and cementite.
- viii) Galvanization is a process of corrosion control of steel.
- ix) Chromium ranges from 16-18 percentages in ferritic stainless steel.
- x) Duralumin is a ferrous alloy.
- xi) Blast furnace is used to produce cast iron.
- xii) Steel with carbon above 0.8% is known as hyper-eutectoid steel.

- 2. a) State the differences between thermosetting and thermoplastic. Give two examples of each type.
b) What is annealing? What is the purpose of annealing process? (2+2)+(4+2)
- 3. a) What is powder metallurgy? Describe sintering process.
b) Differentiate between toughness and resilience.
c) List the applications of low carbon steel. (2+4)+2
- 4. a) What are composite materials? Explain with examples.
b) Distinguish between the full annealing and process annealing?
c) What is case hardening? 4+4+2
- 5. a) Why is cast iron used as a material for machine tool beds?
b) List the properties required for a material to withstand high temperature?
c) Give the composition of High speed steel? 3+5+2
- 6. a) Explain TTT curve for Eutectoid steel? <https://www.wbscteonline.com>
b) What do you mean by Tempering? What are different tempering processes? 5+5
- 7. a) Explain the corrosion mechanism?
b) What are the different methods of preventing corrosion? Briefly explain. 4+6
- 8. a) Name the various methods of manufacturing metal powders and give brief description of each.
b) What is infiltration? Give the metal powders used for porous bearings. 5+5
- 9. a) What do you mean by Engineering materials? Classify the various Engineering materials?
b) What is fatigue? How fatigue failure occurs? 7+3
- 10. a) Describe the types of rubber you know. Explain briefly.
b) Give some typical applications of Thermoplastic elastomers. 6+4

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December 2019

ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 70***Answer to Question No.1 is compulsory and to be answered first.****This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.****Answer any five questions from the rest.**

1. Fill in the blanks (any twenty):

1 X 20

- i) The creep is that property of a material by virtue of which it undergoes a _____ and _____ deformation at constant stress.
- ii) Grey cast iron is _____ than white cast iron.
- iii) The iron ore mostly used for the production of pig iron is _____.
- iv) The plasticity is that property of a material by virtue of which it deforms _____ under the action of external force.
- v) The creep is that property of a material by virtue of which it undergoes a _____ and _____ deformation at constant stress.
- vi) The ultimate stress of a material is _____ than the breaking stress.
- vii) Cutting tools are made from _____ steel.
- viii) Grey cast iron is _____ than white cast iron.
- ix) The iron ore mostly used for the production of pig iron is _____.
- x) Ball bearing are usually, made from _____ steel.
- xi) An alloy of copper, tin and zinc is known as _____.
- xii) German silver consists of _____.
- xiii) Babbit metal is a _____ based alloy.
- xiv) The metal suitable for bearing subjected to high load is _____.
- xv) Brass is an alloy of _____ and _____.
- xvi) The heat treatment process used for softening hardened steel is _____.
- xvii) The process used for relieving the internal stresses previously set up in the metal and for increasing the machinability of steel is _____.
- xviii) Carburizing is a _____ process.

- xix) Normalizing _____ the mechanical properties of steel.
- xx) Normalizing of steel is done to _____ the grain structure.
- xxi) Phenolic resins are properly known as _____.
- xxii) The property of a material by virtue of which it can withstand or support an external force or load without rupture, is known as _____.
2. a) What do you mean by heat treatment of steel?
b) Describe any two of the following heat treatment processes – i) cyaniding, ii) carburising, iii) induction hardening, iv) nitriding. 4+6
3. a) Write a short note on corrosion.
b) What is the difference between creep and fatigue?
c) Write a short note on – i) pitting, ii) erosion. 3+3+4
4. a) Define alloy steel. Explain the effect of nickel, chromium and manganese on the properties of steel.
b) Describe aluminium casting alloys and Y-alloy in brief. 6+4
5. Describe: blending, compacting, pre-sintering and sintering method. 10
6. a) Write briefly about the different stages of powder metallurgy.
b) State the advantages of powder metallurgy. 8+2
7. a) How do thermo plastics differ from thermosetting plastics?
b) Write short notes on – i) Thermocole, ii) Glass wool.
c) State the properties of refractory materials. 3+4+3
8. a) Draw iron carbon equilibrium diagram and write the name of different phases and mention different.
b) What is the Duralumine? 7+3
9. a) State the properties and applications of copper alloy.
b) What is bearing material? 7+3
10. a) Difference between destructive testing and non destructive testing.
b) Write a short note on – i) Dry penetrant test, ii) Magnaflux test. 5+5

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MECHANICAL ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten): 1 x 10
- i. When a steel is heated in a furnace and then cooled in air at ordinary temperature, the process is
(a) Annealing (b) Hardening (c) Normalizing (d) Tempering
 - ii. γ form of iron is?
a) Magnetic b) Non magnetic c) Occurs below 768°C d) Occurs above 1410°C
 - iii. Bronze is an alloy of
(a) copper & zinc (b) copper & tin (c) copper, zinc & tin (d) any one of these
 - iv. Fatigue failure can be detected with the help of – a) stress – strain curve, b) S – N curve, c) graph of law of machine, d) iron – carbon diagram.
 - v. In low carbon steel C % varies from- (a) 0.008-0.3 (b) 0.3-0.6 (c) 0.6-2.0 (d) 2.0-4.3
 - vi. In the austempering process of heat treatment, austenite changes into
a) martensite b) troostite c) sorbite d) bainite
 - vii. Gold has the crystal structure of- (a) BCC (b) FCC (c) HCP (d) RCC
 - viii. The name diffusion less transformation is associated with –
a) pearlite, b) bainite, c) martensite, d) cementite.
 - ix. Silicon when added to copper improves-
(a) machinability, (b) hardness, (c) hardness and strength, (d) strength and ductility
 - x. The ability of a material to be drawn into thin wire is known as-
a) Malleability b) Ductility c) Harness d) Brittleness
 - xi. The temperature (in °C) of peritectic reaction in iron-carbon phase diagram is-
a) 1539 b) 1493 c) 910 d) 1175
 - xii. The machine tool guide ways are usually hardened by
(a) vacuum hardening, (b) martempering, (c) induction hardening, (d) flame hardening
 - xiii. The hardenability is not affected by -
(a) air, (b) chemical composition of steel, (c) critical cooling rate, (d) quenching medium and method of quenching
 - xiv. To obtain martensite structure – a) slow cooling is to be done, b) quick cooling is to be done, c) air cooling is to be done, d) the cooling curve must cross C- curve.
 - xv. Stainless steel is so called because of its-
(a) High strength (b) High corrosion resistance
(c) High ductility (d) Brittleness

2. Fill in the blanks (any ten):

1x10

- a) The number of effective atoms in a HCP unit cell is _____.
- b) The ultimate stress of a material is _____ than the breaking stress.
- c) Plastic cables are made by _____ process.
- d) _____ metal has the highest electrical conductivity.
- e) Curie temperature for iron is _____.
- f) The iron ore mostly used for the production of pig iron is _____.
- g) Carbon percentage in cast iron is _____.
- h) Most metals has _____ structure.
- i) Y-alloy is an important alloy of _____.
- j) The steel used for making shafts is _____.
- k) Babbit metal is a _____ based alloy.
- l) _____ as a medium is used for fastest cooling rate of steel quenching.
- m) The cupola is used to manufacture _____.
- n) BUNA- N is also called _____.
- o) The heat treatment process used for softening of hardened steel is _____.

3. Answer the following questions (any ten): <https://www.wbscteonline.com>

1x10

- a) Why polymer cannot be remoulded?
- b) What is the full form of PVC?
- c) What is creep?
- d) What is the last step in magnetic particle test method?
- e) What is 18:4:1 tool steel?
- f) Give two examples of Thermoplastics.
- g) Write two applications of Muntz metal.
- h) Give two examples of non destructive testing.
- i) Briefly state the application of Inver.
- j) What is the unit of strain?
- k) What is unit cell?
- l) What is meant by work hardening?
- m) What is the difference between bronze and brass?
- n) What does ASTM stand for?
- o) What is radiography test?

Group B

4. Answer the following questions (any six):

2x6

- i. What are the desirable properties of bearing materials?
- ii. How plain carbon steel is classified?
- iii. What are composite materials?
- iv. Write down the difference between hardness & toughness.
- v. What is Duralumin?
- vi. Write the composition of '40 Ni 2 Cr 1 Mo 25' steel.
- vii. What is an 'Allotropic metal'?
- viii. What is powder metallurgy process?
- ix. What is the purpose of annealing?
- x. What is phase diagram?

Group C

5. Answer the following question (any one):

1x6

- a) Draw the 'Iron-carbon equilibrium diagram' and explain it. Briefly describe Surface hardening and tempering as heat treatment processes.

4+2

- b) Determine the average number of atoms, relation between lattice parameter and atomic radius, atomic packing factor of BCC crystal structure. 2+2+2
- c) What do you mean by a unit cell? Volume of a unit cell and one atom of the unit cell are 100 nm^3 and 10 nm^3 respectively. If the unit cell has 5 numbers of effective atoms then find out the atomic packing factor (APF) of the unit cell. 2+4
6. Answer the following question (any one): 1x6
- a) Explain the necessity of heat treatment for steel. Describe the process of quenching. 4+2
- b) Write short notes on - (i) induction hardening, (ii) flame hardening 3+3
- c) Describe normalizing process. State its purpose. 4+2
7. Answer the following question (any one): 1x6
- a) State two differences between thermosetting plastics and thermoplastics with proper examples. Give one application for each of the followings- (a) ABS, (b) Acrylics, (c) Bakelite. 3+3
- b) Differentiate between destructive testing and non destructive testing. Write short note on – i) Dry penetration test, ii) Magnaflux test. 2+2+2
- c) What is a polymer ? Give two examples. What is Vulcanized Rubber? 2+2+2

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MECHANICAL ENGINEERING MATERIALS*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): (1x20)
 - i) Coordination number FCC structure is - (a) 6 (b) 8 (c) 10 (d) 12.
 - ii) Pure iron at room temperature has the following crystal structure. - (a) BCC (b) FCC (c) HCP (d) SC.
 - iii) Atomic packing factor of HCP structure is - (a) 0.74 (b) 0.68 (c) 0.52 (d) none of these.
 - iv) The property by virtue of it a material can be rolled into thin sheet is called a) Ductility b) elasticity c) malleability d) Fatigue.
 - v) The continuous deformation under steady load is called - (a) creep (b) fatigue (c) corrosion (d) none of these.
 - vi) Hardest micro structure of steel is - (a) Ferrite (b) Martensite (c) Bainite (d) Pearlite.
 - vii) The steel having more than 0.8% carbon are called - (a) eutectoid steels (b) hypo eutectoid steel (c) hyper eutectoid steel (d) none of these.
 - viii) The property of material by virtue of which it undergoes permanent deformation with the application of a load, is called - (a) hardness b) plasticity c) Fatigue d) elasticity.
 - ix) In Annealing heat treatment cooling is done in (a) air (b) Furnace (c) water (d) None of these
 - x) Product of blast furnace is - (a) Wrought Iron (b) Cast Iron (c) Pig Iron (d) Steel
 - xi) In low carbon steel C % varies from - (a) 0.008-0.3 (b) 0.3-0.6 (c) 0.6-2.0 (d) 2.0-4.3
 - xii) Cast iron is a - (a) ductile material b) malleable material c) brittle material d) tough material.
 - xiii) Which of the following iron exist at room temperature? - (a) alpha Iron (b) Beta Iron (c) gamma Iron (d) delta Iron
 - xiv) The hardness of steel increases if % of carbon (a) decreases (b) Increases (c) all of the above (d) None of the above
 - xv) Tempering heat treatment is done after - (a) Hardening, (b) normalising, (c) Annealing, (d) None of the above.
 - xvi) Which is surface hardening process? a) Annealing b) Normalising c) Nitriding d) hardening
 - xvii) The lower critical point for all steel is- (a) 600°C (b) 723°C (c) 700°C (d) 913°C.
 - xviii) Brass is an alloy of - (a) copper and zinc (b) copper and tin (c) copper, Aluminium (d) none of these.

- xix) Duralumin is an alloy of – a) Aluminium b) iron c) Nickel d) None of the above
- xx) Babbit metal is used in – (a) Bearing (b) Gear (c) Shaft (d) None of these
- xxi) Bottles from thermoplastic materials are made by – (a) Compression moulding b) Extrusion moulding c) Injection moulding d) blow moulding.
- xxii) Long plastic pipes are produced by – (a) Compression moulding b) extrusion moulding c) Injection moulding d) blow moulding.
- xxiii) Examples of Thermosetting plastics are – (a) Electric switch (b) Pressure cooker handle c) all of the above d) None of the above
- xxiv) Example of non destructive test is- (a) ultrasonic test (b) radiographic test (c) magnetic particle test (d) all of these.
- xxv) Non- destructive testing is used to determine a) Location of defects, b) Chemical composition, c) Corrosion of metal, d) All of these.

Group B

2. a) Write down the names of different types of crystal structures in metals. (2)
b) Write down the number atoms per unit cell, co ordination number, atomic packing factors and relation between sides of unit cell and atomic radius of simple cubic and BCC structure. (6)
3. a) Define hardness, Brittleness, ductility and Elasticity. (4)
b) Draw schematic diagram showing flow path for making steel and cast iron, wrought iron in different furnaces from iron ore. (4)
4. a) Draw the cooling curve for pure iron showing salient points. (2)
b) Draw Iron carbon equilibrium diagram up to 2 % carbon showing micro constituent present in (4)
c) Classify steel according to % of carbon. <https://www.wbscteonline.com> (2)
5. a) Explain the various purpose of heat treatment. (3)
b) Describe hardening heat treatment process and also write down its application. (5)
6. a) Write short note on any one i) Nitriding ii) Flame hardening iii) Austempering (4)
b) Draw the TTT diagram for eutectoid steel showing salient points. (4)
7. a) Write down the names of two copper based alloys. (2)
b) What are the properties of aluminium alloys? (2)
c) Describe briefly the properties and application of bearing metals. (4)
8. a) Compare Thermoplastics and Thermosetting plastic. (3)
b) Describe Compression moulding or Injection moulding process with the help of simple sketch. (5)
9. Write short notes on any two i) Fibre reinforced Composite Material ii) Ceramics iii) Bakelite
iv) refractories v) polyester (4+4)
10. a) Write down the names of different Non destructive testing methods. (2)
b) What are the differences between destructive and non destructive testing methods? (2)
c) Describe any one non destructive testing method with suitable sketch. (4)



DECEMBER 2024

MECHANICAL ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from the rest.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (Any ten): 2x10=20

- i) An alloy of copper and zinc is known as- (a) Brass (b) Monel metal (c) Bronze (d) Hindalium.
- ✓ ii) German silver contains- (a) 70% Cu, 30% Zn (b) 60% Cu, 40% Zn (c) 40% Zn and 1% tin (d) 60.45% Cu, 35.2% Zn, 5.35% Ni.
- ✓ iii) When the austenitic steel (above 723°C) is cooled in air, then the resulting structure will be- (a) Fine pearlite (b) Martensite (c) Coarse pearlite (d) None of these.
- iv) Which alloying element is added to cast iron mainly with a view to mitigate the ill effects of sulphur?
(a) silicon (b) phosphorus (c) manganese (d) molybdenum.
- ✓ v) Which of the following is basically used in forging purpose? (a) Medium carbon steel (b) High speed steel (c) High carbon steel (d) Mild steel.
- ✓ vi) Which of the following alloy is mainly used for aircraft application? (a) Bronze (b) Babitt metal (c) Duralumin (d) None of these.
- vii) Fibre reinforced plastic is- (a) alloy (b) ceramic (c) composite (d) metal.
- viii) The property which is desirable in a material subjected to shock loading – (a) Ductility (b) Malleability (c) Hardness (d) Toughness.
- ix) Cast iron is a product of- (a) Cupola (b) Bessemer converter (c) Open hearth furnace (d) Blast furnace.
- x) Red hot hardness of an alloy steel can be improved by adding - (a) Tungsten (b) Cobalt (c) Manganese (d) Both (a) & (b).
- xi) Which of the following is the hardest? (a) Austenite (b) Pearlite (c) Bainite (d) Martensite.

- xii) Which of the following is/are an example of Thermo-setting plastic? (a) Melamine (b) Epoxies (c) Bakelite (d) All of these.
- xiii) Low cost camera lenses are made by - (a) Acrylics (b) Nylon (c) ABS (d) None of these.
- xiv) The smallest part of a crystal lattice is known as - (a) Lattice structure (b) Lattice point (c) Bravais lattice (d) Unit cell.
- xv) The lower critical temperature (in °C) in iron-carbon phase diagram is - (a) 1539 (b) 1143 (c) 910 (d) 723.

GROUP-B

- 2) (a) Why tempering is required in Heat Treatment operation?
(b) Explain Martempering process with suitable TTT diagram. 2+6
- 3) (a) Classify Plain carbon Steels along with their composition and applications.
(b) Write down the composition and use of wrought iron. 6+2
- 4) (a) Write down the advantages, limitations of Ultrasonic Crack Detection and its area of application. <https://www.wbscteonline.com>
(b) State the properties and application of ABS and polyesters. (4+4)
- 5) (a) What is the meaning of 'XT 72 W 18 Cr 4 V1'?
(b) Identify different alloying elements along with their effect on steel. (2+6)
- 6) (a) State the common properties of plastics.
(b) Explain briefly the working principle of Plastic Injection moulding process with neat sketch. (2+6)
- 7) (a) Describe Allotropy of Iron.
(b) Draw Iron-Carbon equilibrium diagram indicating different phases, their carbon percentage and temperature of transformation. (2+6)
- 8) (a) Write down the property, composition and use of Y-alloy and Gun metal.
(b) Describe Flame Hardening process. (5+3)
- 9) (a) Write down the Coordination Number and Atomic Packing Factor for SC, BCC and FCC.
(b) Specify two important methods to prevent the corrosion. (6+2)

MECHANICAL ENGINEERING MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20
- Which of the following material has maximum ductility- i) Mild steel ii) Copper iii) Nickel iv) Aluminium
 - ☒ An eutectoid steel consists of- i) wholly pearlite ii) wholly austenite ☒ iii) pearlite and ferrite iv) pearlite and cementite
 - ☒ The hardness of steel depends upon the- i) amount of cementite it contains ☒ ii) amount of carbon it contains iii) contents of alloying elements iv) method of manufacture of steel
 - Blast furnace is used to produce- i) ☒ pig iron ii) steel iii) wrought iron iv) cast iron
 - The strength is the ability of a material to resist- i) ☒ deformation under stress ii) externally applied forces with breakdown or yielding iii) fracture due to high impact loads iv) none of these
 - ☒ In induction hardening _____ is high. i) ☒ current ii) voltage iii) frequency iv) temperature
 - Which of the following is an amorphous material- i) Mica ii) Glass iii) Lead iv) Silver
 - ☒ The type of space lattice found in alpha-iron is- i) FCC ☒ ii) BCC iii) HCP iv) None of these
 - i) The toughness of a material _____ when it is heated. i) remains same ☒ ii) increase iii) decrease
 - j) Muntz metal (Yellow brass) contains i) 70% copper and 30% zinc ii) 60% copper ☒ and 40% zinc iii) 59% copper, 40% zinc and 1% tin iv) 60.45% copper, 35.2% zinc and 5.35% nickel
 - ☒ Vanadium when added to steel- i) decreases tensile strength ii) raises critical temperature iii) Sowers critical temperature ☒ iv) increases tensile strength
 - ☒ i) increases tensile strength i) increases tensile strength ii) increases tensile strength iii) increases tensile strength ☒ iv) increases tensile strength
 - m) Which of the following property is desirable in parts subjected to shock and impact loads i) ☒ Strength ii) Brittleness iii) Stiffness iv) Toughness
 - n) The value of Atomic packing factor for BCC crystal structure is- i) 0.68 ii) 0.74 ☒ iii) 0.29 iv) none of these
 - ☒ o) Monel metal contains- i) 65% nickel, 15% chromium and 20% iron ii) ☒ 68% nickel, 29% copper and 3% other constituents iii) 80% nickel and 20% chromium iv) 80% nickel, 14% chromium and 6% iron

GROUP-B

2. ☒ a) Write short notes on - a) Bainite b) Spheroidite c) Martensite 6
☒ b) Write down few differences between Case Hardening & Flame Hardening 2
3. ☒ a) Write down composition of a) Munz metal b) Gun Metal c) Duralumin d) Y-Alloy 4
☒ b) What is Corrosion? Name few corrosion resistive materials? Mention two corrosion preventive methods. 4
4. ☒ a) Determine the average number of atoms, relation between lattice parameter and atomic radius, atomic packing factor of FCC crystal structure. 2+2+2
☒ b) Describe the process of Nitriding. 2

- | | | | |
|----|----|--|-------|
| 5. | a) | Give one application for each of the followings - a)Acrylics b)Polyesters c)Epoxies d)Melamine | 4 |
| | b) | Briefly explain Dye penetration test. | 4 |
| 6. | a) | Describe Annealing process. State it's Purpose. | 4+2 |
| | b) | What is Martempering? | 2 |
| 7. | a) | Briefly explain a) Eutectic reaction b) Eutectoid reaction c) Peritectic reaction | 2+2+2 |
| | b) | Write down differences between White cast iron and Grey cast iron. | 2 |
| 8. | a) | Write down composition of Duralumin and Magnalium. | 2 |
| | b) | Write down one application each for laminated materials and fibre reinforced materials. | 2 |
| | c) | What do u mean by Vulcanization of Rubber? State its application. | 2+1 |
| 9. | a) | Write down differences between Hardness and Toughness. | 2 |
| | b) | What is powder metallurgy process? Write down few application of this process | 2+1 |
| | c) | Classify plain carbon steel and also mention their respective carbon percentage. // | 2+1 |

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MECHANICAL ENGINEERING MATERIALS

318/2(N)

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

- i) ☒ Coordination number of FCC structure:
 a) 6 b) 8 c) 10 ☒ 12
- ii) ☒ Pure iron at room temperature has:
 a) BCC b) FCC c) HCP d) SC
- iii) ☒ Atomic packing factor (APF) of BCC:
 a) 0.74 ☒ 0.68 c) 0.52 d) None of these
- iv) ☒ Which property enables a material to resist scratching, cutting and abrasion:
 ☒ Hardness b) Plasticity c) Fatigue d) Elasticity
- v) ☒ Toughness of a material when heated:
 a) Remains same ☒ Increases c) Decreases d) none of these
- vi) ☒ Hardest microstructure of steel:
 a) Ferrite ☒ Martensite c) Bainite d) Pearlite
- vii) ☒ Hardness of steel increases when % C:
 a) Decreases ☒ Increases c) All of the above d) None of the above
- viii) ☒ An eutectoid steel consists of:
 ☒ Wholly pearlite b) Wholly austenite c) Pearlite & ferrite d) Pearlite & cementite
- ix) ☒ In the Austempering process, austenite changes into:
 a) Martensite b) Troostite c) Sorbite ☒ Bainite
- x) ☒ In annealing, cooling is done in:
 a) Air ☒ Furnace c) Water d) None of these
- xi) ☒ Tempering is done after:
 ☒ Hardening b) Normalising c) Annealing d) None of these
- xii) ☒ Lower critical point for steels ($^{\circ}\text{C}$):
 a) 600 ☒ 723 c) 700 d) 913
- xiii) ☒ Babbitt metal is used in:
 ☒ Bearings b) Gears c) Shafts d) None of these
- xiv) ☒ Long plastic pipes are produced by:
 a) Compression moulding ☒ Extrusion moulding c) Injection moulding d) Blow moulding
- xv) ☒ NDT is used to determine:
 ☒ Location of defects b) Chemical composition c) Corrosion of metal d) All of these

GROUP-B

Answer any Five (05) questions.

- 2.(a) What do you mean by a unit cell? (2)
- (b) For a Body-Centered Cubic (BCC) crystal structure, determine: (2+4)
 - (i) Average number of atoms per unit cell
 - (ii) Atomic packing factor

3. Define the following mechanical properties: i) Elasticity ii) Ductility iii) Malleability iv) Fatigue (2+2+2+2)
4. Draw the Iron-Carbon phase diagram and label all important points. (8)
5. (a) Write a short note on Cementite and Pearlite. (3 + 3)
 (b) Write the composition of High-Speed Steel (HSS). (2)
6. (a) Draw the TTT diagram for eutectoid steel showing all salient points. (5)
 (b) Write short notes on Case hardening (3)
7. (a) Describe the Annealing process. (4)
 (b) Write the purpose of Austempering and Martempering processes. (2 + 2)
8. (a) Write down the names of two copper-based alloys. (2)
 (b) Write the composition of Muntz Metal and Gun Metal. (2 + 2)
 (c) What are the applications of Monel Metal? (2)
9. (a) Compare Thermoplastics and Thermosetting plastics. (3)
 (b) With the help of a diagram, describe the Injection Moulding process. (5)
10. (a) What are the differences between Destructive and Non-Destructive Testing? (3)
 (b) Explain the Dye Penetration Test with a suitable sketch. (5)



MANUFACTURING PROCESSES-I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$
 - i. The hot working of metals is carried out -
 - a) At the recrystallization temperature
 - b) Below the recrystallization temperature
 - c) Above the recrystallization temperature
 - d) At any temperature
 - ii. Mechanical property desired for successful forging is - (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness.
 - iii. In a four high rolling mill, there are four rolls out of which -
 - a) One is working roll and three are backing up rolls
 - b) Two are working rolls and two are backing up rolls
 - c) Three are working rolls and one is backing up roll
 - d) All of the four are working rolls
 - iv. Seamless tubes in mass production are manufactured by - (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - v. Blanking and punching operations can be performed simultaneously on -
 - (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - vi. Lathe bed is made of a) Mild Steel b) High Carbon steel c) Cast Iron d) High Speed Steel
 - vii. The angle between face and flank of a single point cutting tool is known as a) Rake Angle b) Clearance Angle c) Lip Angle d) Nose Angle
 - viii. Which is an example of orthogonal cutting? (a) Milling, (b) Turning (c) Drilling, (d) Planning.
 - ix. Which is related with offsetting the tail stock? a) Turning b) Facing c) Taper turning d) Knurling.
 - x. The allowance given for easy removal of pattern is a) Shrinkage allowance b) Wrapping Allowance c) Machining Allowance d) Draft Allowance
 - xi. Gate is provided in moulds to - (a) Feed the casting at a constant rate, (b) Give passage to gases, (c) Compensate for shrinkage, (d) Avoid cavities.
 - xii. Metal and alloys of which following metal is used for cold chamber die casting-(a) tin, (b) lead, (c) aluminium, (d) iron
 - xiii. The flame used for welding brass is - a) Oxidising b) Neutral c) Carburising d) Natural Flame
 - xiv. The welding arc is not visible to worker in a) MMAW b) GTAW c) GMAW d) SAW
 - xv. Spot welding is one of the type of a) Arc Welding b) Resistance Welding c) Pressure Welding d) Solid-Liquid State welding

GROUP-B

Answer any Five (05) questions.

2. i) What are the advantages of hot working over cold working process?
ii) With suitable diagram explain the open die forging process. 4+4
 3. i) What do you mean by rolling? Discuss 2 Hi and 3 Hi roll stand arrangement with sketches.
ii) Name two forging defects and state the reason for arising these defects. (1+5)+2
 4. i) Discuss forward extrusion and backward extrusion with neat sketches.
ii) Describe sheet metal operations (any two) with sketch. a) Blanking, b) Punching c) Drawing. 4+4
 5. i) State the desirable properties of cutting tool materials.
ii) Explain the different types of chips formed during machining with sketches. 2+6
 6. i) Differentiate between orthogonal cutting and oblique cutting.
ii) Define Cutting speed, feed and depth of cut in Machine tool.
iii) State the functions of cutting fluid. 3+3+2
 7. i) Explain the nomenclature of a single point cutting tool with a neat sketch.
ii) Metric thread of 0.8 mm pitch is to be cut on a lathe. Pitch of the lead screw is 1.5 mm. Find the change gear ratio. 6+2
 8. i) Draw and label the elements of a gating system and briefly explain the function of each.
ii) Explain the causes and remedies of three casting defects with suitable sketches. 5+3
 9. i) Describe the different types of pattern allowances in foundry work.
ii) State the difference between MIG welding and TIG welding. 5+3
 10. i) Describe the characteristics and applications of different gas welding flames with neat sketches.
ii) Distinguish between brazing and soldering. 6+2
-

MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70*

Answer to Question No.1 is compulsory and to be answered first.
 This answer is to be made in separate loose script(s) provided for the purpose.
 Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and
 fresh answer scripts for answering the remaining part of the question will be provided.
 On early submission of answer scripts of Question No.1,
 a student will get the remaining script earlier.

Answer any five questions from the rest.

1. A. Fill in the blanks (any ten): 1x10
- i) The ratio of _____ and acetylene in neutral flame is 1:1.
 - ii) Electrode material for _____ welding is Tungsten.
 - iii) _____ flame is used for welding brass.
 - iv) Axis of the casting and axis of rotation is different in _____ casting process.
 - v) In _____ die-casting method continuous heating of metal is not necessary.
 - vi) _____ allowance is given to compensate effect after solidification.
 - vii) _____ allowance is given for easy removal of pattern.
 - viii) Master pattern is used for making wax pattern for _____ casting method.
 - ix) In _____ welding process arc is not visible to welder.
 - x) Seam welding may be considered as continuous _____ welding.
 - xi) _____ welding is a modification of spot welding.
 - xii) Core box is used to make _____.
 - xiii) Lathe is specified by _____ of the job over carriage.
 - xiv) A small depression in the parent metal during arc welding is called _____.
- B. Choose the correct answer (any ten): 1x10
- i) Solid / split / sweep pattern is made in a single piece.
 - ii) Spot / TIG / Gas welding is an example of resistance welding.
 - iii) Spot / Seam / TIG welding is used in automobile industry.
 - iv) Electrode is not consumed during TIG / MIG / spot welding.
 - v) Projection / Gas / TIG is an example of resistance welding.
 - vi) Spelter is used during soldering / brazing / TIG.
 - vii) Spot welding electrode is made off copper / aluminium / iron.
 - viii) Brazing is similar to soldering / Gas welding / oxy-cutting.
 - ix) Skeleton pattern is used when pattern is big / small.
 - x) Casting is a forming / deforming process.
 - xi) Sanding is a surface finishing / machining process.
 - xii) Brazing is a joining process / removing process.
 - xiii) The filler material used during soldering is called solder / spelter.
 - xiv) Segmental pattern is used for circular / square work.
2. Describe hot chamber and cold chamber die-casting method.
3. Write short notes on – (a) Sweep pattern, (b) Loose-piece pattern. 5+5
4. Describe submerged arc welding process. 5+5

5. What is core print? What is core? Write the advantages of wood as pattern making material. 5+5
 6. How a Lathe machine can be specified? Write the functions of Lathe machine parts? 5+5
 7. Write the difference between up milling and down milling. Write the specifications of drilling machine. 5+5
 8. Write shorts on – (a) Facing, (b) Parting-off, (c) Knurling. 3+3+4
 9. Describe different types of milling cutter. 10
 10. Write short notes on different types of chips. 10
 11. Write the functions of different parts of a radial drilling machine. 10
-

December 2018

MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70***Answer to Question No.1 is compulsory and to be answered first.****This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.****Answer any five questions from the rest.**

- I. Answer the following questions (any twenty): 1x20
- i) Mechanical property desired for successful forging is – (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness
 - ii) Seamless tubes in mass production are manufactured by – (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - iii) In drop forging, forging is done by dropping – (a) The work piece at high velocity, (b) The hammer at high velocity, (c) The die with hammer with high velocity, (d) A weight on hammer to produce the requisite impact.
 - iv) Blanking and punching operations can be performed simultaneously on – (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - v) Fluxes are used while welding – (a) To increase the rate of welding, (b) To clean the joint, (c) To prevent oxidation of metal during welding, (d) All the above.
 - vi) Oxy-acetylene flame is used for welding of – (a) Monel, (b) Nickel, (c) Steel, (d) Bronze.
 - vii) The hidden defect of a weld can be detected by – (a) Tensile test, (b) Bend test, (c) Radiography, (d) All.
 - viii) Commonly used gases in TIG welding are – (a) Hydrogen & Oxygen, (b) Hydrogen & Argon, (c) Hydrogen & Helium, (d) Helium & Argon.
 - ix) Consumable electrode is used in the following process – (a) TIG welding, (b) Gas welding, (c) Arc welding, (d) Resistance welding.
 - x) Above _____ temperature, working of metals is named hot working.
 - xi) Drawing is done to _____ the diameter of wire.
 - xii) In 4-high rolling mill, bigger rolls are known as _____ rolls.
 - xiii) Accurate taper and step tapers can be machined with _____ attachment.
 - xiv) Finishing and sizing a hole is called _____.
 - xv) A twist drill has _____ flutes.

- xvi) Pattern used in precision investment casting normally made of _____.
- xvii) In resistance welding _____ is generated by the resistance offered to the flow of current.
- xviii) MIG welding uses _____ electrode.
- xix) In neutral flame in Oxyacetylene gas welding, ratio of acetylene to oxygen is _____.
- xx) Spot welding is one of the _____ welding processes.
- xxi) The revolving cutter of a milling machine is held on a _____.
- xxii) In universal milling machine table can have the _____ arrangement about a vertical axis.
-
2. a) What is Pattern? Name different types of pattern. Sketch any three of them and explain.
 b) What are the pattern making allowances? Explain any one of them. 7+3
3. a) Discuss with sketches hot rolling and cold rolling.
 b) Discuss with neat sketches direct extrusion and indirect extrusion. 5+5
4. a) How do you specify a Lathe machine?
 b) Write short notes on orthogonal cutting and Oblique cutting.
 c) What are the different operations that can be done on a lathe machine? 3+5+2
5. a) How do you specify a Drilling machine? Draw a pillar type Drilling machine & mention basic parts.
 b) Write short notes on Up Milling & down Milling. 7+3
6. a) What is core? What is the purpose of core?
 b) Explain the defects in casting. How can those defects be prevented? 5+5
7. a) Explain Arc welding process showing all salient points on it.
 b) State various welding defects & explain the remedies. 5+5
8. a) Explain spot welding and show its simple set up.
 b) Name the different flames and show the different zones in each of them. 5+5
9. Write short notes on the followings (any four): 2½x4
 (a) Knurling, (b) Reaming, (c) Brazing, (d) Soldering, (e) Core Prints, (f) Centrifugal Casting, (g) Die Casting
-

MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70*

Answer to Question No.1 is compulsory and to be answered first.

**This answer is to be made in separate loose script(s) provided for the purpose.
Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.**

**On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.**

Answer any five questions from Group-A, B & C, taking at least one from each group.

1. Answer the following questions (any twenty): 1x20
 - i) In Oxyacetylene gas welding with neutral flame, the ratio of acetylene to oxygen is – (a) 1:2, (b) 1:1, (c) 2:1, (d) 1:3.
 - ii) Non-consumable electrode is used in the – (a) MIG welding, (b) TIG Welding, (c) Arc welding, (d) Gas welding.
 - iii) Chances of crack propagation are more in – (a) cold working process, (b) hot working process, (c) both (a) & (b), (d) none of these.
 - iv) The gating ratio 2:8:1 for copper in gating system design refers to the ratio of areas of – (a) sprue : runner : ingate, (b) runner: ingate : sprue, (c) runner : sprue : ingate, (d) ingate : runner : sprue.
 - v) The centre lathes receive their power through – (a) headstock, (b) tailstock, (c) both (a) & (b), (d) none of these.
 - vi) In oblique cutting system, the maximum chip thickness – (a) occurs at the middle, (b) may not occur at the middle, (c) depends upon the material of the tool, (d) depends upon the geometry of the tool.
 - vii) Which current is used in Tungsten Inert-Gas (TIG) welding? – (a) Only A.C. can be used as welding current, (b) Only D.C. can be used as welding current, (c) Both A.C. and D.C. can be used as welding current, (d) None of these.
 - viii) Gate is provided in moulds to – (a) Feed the casting at a constant rate, (b) Give passage to gases, (c) Compensate for shrinkage, (d) Avoid cavities.
 - ix) A moving mandrel is used in – (a) wire drawing, (b) tube drawing, (c) metal cutting, (d) forging.
 - x) Scab is a – (a) welding defect, (b) sand casting defect, (c) machining defect, (d) forging defect.
 - xi) Which of the following gas mixtures is not used in Gas Tungsten arc welding (TIG)? – (a) Argon-Helium, (b) Argon-Nitrogen, (c) Argon-Hydrogen, (d) Argon-CO₂.
 - xii) Which of the following is always a cold working operation? – (a) Rolling, (b) Wire drawing, (c) Forging, (d) Extrusion.
 - xiii) The angle between face and flank of a single point cutting tool is known as _____.
 - xiv) In _____ welding process, the arc is not visible to worker.

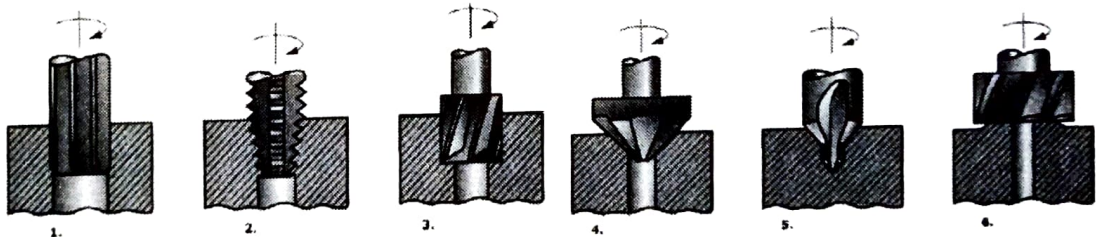
- xv) _____ is a mixture of finely divided aluminum and iron oxides.
- xvi) Cuts and washes in sand casting is caused by _____ of molding sand by the flowing metal.
- xvii) Spot/Seam/TIG welding is used in automobile industry.
- xviii) Hot working of metal in metal forming processes is working _____ recrystallization temperature.
- xix) In MIG welding, _____ current _____ polarity is generally used.
- xx) Single point cutting tool should ideally have _____ rake angle.
- xxi) A lathe bed is generally made of _____.

Group-A

- 2. a) What do you mean by forming? What is recrystallization?
 b) Distinguish between forward extrusion and backward extrusion with neat sketches. (1+2)+4+3
 c) What are the advantages of hot working over cold working process?
- 3. a) What do you mean by rolling? Discuss different kinds of roll stand arrangement.
 b) State three forging defects and state the reason for arising these defects. (1+4)+2+3
- 4. a) Discuss different stages of drop forging.
 b) What is the difference between punching and blanking? What is meant by stripping force and punching force? 5+5

Group-B

- 5. a) Differentiate between orthogonal cutting and oblique cutting.
 b) Explain the nomenclature of a single point cutting tool with a neat sketch. 4+6
- 6. a) Differentiate between Capstan and Turret lathe.
 b) Identify the following machining operations and define them. 4+6



- 7. a) How do you specify a milling machine?
 b) Differentiate between conventional milling & climb milling.
 c) Write short notes on any two – (a) Knurling, (b) facing, (c) Parting off. 2+4+4
- 8. a) What are the different types of chips formed during machining? Write short note on Continuous chip with built-up edges.
 b) Draw the neat sketch of a twisted drill bit and mark various part of it. (1+3)+6

Group-C

- 9. a) Define core print. What is the use of it?
 b) Write down the advantages and disadvantages of green sand mould.
 c) Write short notes on – (i) Sweep pattern, (ii) Loose-piece pattern. 2+4+4

10. a) Draw the diagram depicting two-piece sand casting arrangement and show the locations of pouring basin, sprue, gate, runner, riser, mould cavity, parting line, air vent, cope and drag on it.
b) Name three welding defects. State the reason for arising any one of such defects and the remedies to avoid them. 5+(1+2+2)
11. a) What are the typical electrode materials used for Tungsten Inert gas welding?
b) Explain the characteristics of Arc welding machines, which viz., constant current and constant voltage. Mention the applications of each of them.
c) What are the problems encountered with the use of coated electrodes? Explain how these can be taken care of. 2+4+4
12. a) Distinguish between – (i) hot chamber and cold chamber die casting, (ii) Soldering & Brazing.
b) Explain precision investment casting process with neat sketch. (3+3)+4
-



DECEMBER 2024

3236244**MANUFACTURING PROCESS - I***Time Allowed: 3 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script.
In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

- (i) The point at which the cutting tool reaches, beyond which it will not function satisfactorily until it is reground, is called as: a) Tool wear, b) Tool failure, c) Tool diffusion, d) None of the above.
- (ii) Which cutting condition affects the cutting temperature predominantly?
 a) Depth of cut, b) cutting speed, c) feed, d) All of the above.
- (iii) In a 4-Hi rolling mill, backup rollers are used to _____
 a) Reduce rolling force, b) Increase stiffness of rolling mill, c) Reduce deflection of working rollers, d) Increase reduction in thickness of job.
- (iv) Shaping of metal by application of compressive force in between two or more dies in order to obtain desire shape is done by- a) Forging, b) Rolling, c) drawing, d) blanking.
- (v) In which of the following forging metal is kept in the lower die?
 a) Open die, b) closed die, c) hold die, d) impression die.



- (vi) The metal forming process in blanking operation is associated with which type of force?
 a) Shear force, b) tensile force, c) compressive force, d) all of the above.
- (vii) The energy requirement in direct extrusion is _____ than indirect extrusion.
 a) More, b) less, c) same, d) no correlation.
- (viii) The surface finish of work piece in hot working is _____ the surface finish in cold working.
 a) Better than, b) worse than, c) same as, d) no correlation.
- (ix) By which material is lathe bed made?
 a) High carbon steel, b) cast iron, c) mild steel, d) copper.
- (x) For which kind of chuck is centering not required?
 a) 4-jaw chuck, b) magnetic chuck, c) 3-jaw chuck, d) mandrel.



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- (xi) Which type of feed is needed in facing operation?
a) Longitudinal, b) cross, c) both longitudinal and cross, d) angular.
- (xii) In neutral flame acetylene and oxygen is in _____ by volume.
a) 1:1 ratio, b) 2:1 ratio, c) stoichiometric ratio, d) ratio.
- (xiii) The ability of the moulding sand to withstand the heat of melt without showing any sign of softening is called as _____.
a) Strength of cohesiveness, b) refractoriness, c) collapsibility, d) flowability.
- (xiv) Which of the following is not a requirement of a good pattern material?
a) It should be light in weight, b) it should have smooth surface, c) it should be easily available, d) it should have low strength to break it and remove easily from mould.
- (xv) AC arc in arc welding is used for welding _____.
a) Hard material, b) highly oxidative material, c) ductile material, d) any material.

GROUP-B

2. (i) Classify manufacturing processes. 3
(ii) What is the main advantage of die forging over impact forging? 2
(iii) Discuss about any two Forging Defects. 3
3. (i) Classify rolling mill as per number of rollers. Draw figures for each of them. 3
(ii) Why smaller roller should be used to decrease contact surface for same deformation? 2
(iii) Write differences between direct and indirect extrusion. 3
- * * * * *
4. (i) Briefly discuss about Different components and their functions of standard die set with a punch. 5
(ii) What are punching and blanking operation? 3
5. (i) Draw three views of a single point cutting tool showing different angles, radius and planes in appropriate views. 3
(ii) Show the specifications of a single point cutting tool having nomenclature 8-10-6-6-6-10-0.2. 2
(iii) Name some cutting tool materials. 1
(iv) Mention the factors affecting tool life. 2
6. (i) Specify a centre lathe. 3
(ii) Why Face Plate and Angle Plates are used in Lathe Machine? 2
(iii) Name any three types of lathe operation showing their schematic representation. 3



3236244

7. (i) Draw different types of oxy-acetylene flame showing different zones. 3
(ii) When reverse polarity is used in arc welding? 2
(iii) What are the causes of using flux in arc welding? 2
(iv) Name any one field of application of TIG welding. 1
8. (i) What is match plate pattern? 2
(ii) Name any two types of pattern allowances. 2
(iii) Name the ingredients of moulding sand with their utility. 2
(iv) Write steps in investment casting. 2
9. (i) Discuss any one type of taper turning operation with figure. 3
(ii) Write down field of application of die and centrifugal casting. 2
(iii) Write down basic differences of brazing and soldering. 2
(iv) On what kind of material spot welding can be done? 1
-



JANUARY 2024

MANUFACTURING PROCESSES - I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of 'Group: A' must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from 'Group: B'.

GROUP: A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): **2x10=20**

- i. The metal is subjected to mechanical working for –
a) Refining grain size b) Changing original block into desired shape c) Controlling the direction of flow lines d) All of these
- ii. The hot working of metals is carried out –
a) At the recrystallization temperature b) Below the recrystallization temperature c) Above the recrystallization temperature d) At any temperature
- iii. Structural sections such as rails, angles, I-beams are made by –
a) Hot rolling b) Hot drawing c) Hot piercing d) Hot extrusion
- iv. In indirect extrusion, which type ram is used?
a) Solid ram b) Hollow ram c) Curved ram d) None of these
- v. In a four high rolling mill, there are four rolls out of which –
a) One is working roll and three are backing up rolls b) Two are working rolls and two are backing up rolls c) Three are working rolls and one is backing up roll d) All of the four are working rolls
- vi. In oblique cutting of metals, the cutting edge of the tool is –
a) Perpendicular to the work piece b) Perpendicular to the direction of tool travel c) Parallel to the direction of tool travel d) Inclined at an angle less than 90° to the direction of tool travel
- vii. Among the following, which one is considered as production purpose Lathe?
a) Center Lathe b) Gap bed Lathe c) Engine Lathe d) Capstan and Turret Lathe
- viii. Which of the following is related with offsetting the tail stock?
a) Turning b) Facing c) Taper turning d) Knurling
- ix. In order to hold an irregular cross-sectional job in a Lathe, which type of chuck is suggested?
a) Three jaw self-centered chuck b) Four jaw independent chuck c) Collet chuck d) Magnetic chuck
- x. The time elapse between two successive grinding of a cutting tool is known as –
a) Machining time b) Tool life c) Set-up time d) None of these
- xi. The main functions of cutting fluids are –
a) Lubrication b) Cooling c) Lubrication and cooling d) None of these
- xii. Which form of metal is used in casting process?
a) Solid b) Liquid c) Gas d) Plastic
- xiii. In which casting process master pattern and wax pattern are used?
a) Continuous casting b) Centrifugal casting c) Investment casting d) Shell casting
- xiv. Among the following, which one is fall under the category of pressure welding?
a) TIG welding b) Arc welding c) Resistance welding d) None of these
- xv. Among the following, which welding process does not require any electric power supply during its execution?
a) TIG welding b) MIG welding c) Thermit welding d) Resistance welding

GROUP: B

2. a) State the difference between hot working and cold working processes.
b) State the advantages and disadvantages of forging process.
3. a) With a suitable diagram explain the working of Four High Rolling Mill.
b) Describe with neat sketch the working principle of Direct Extrusion Process.

[4+4]

[4+4]

4. a) Define Tool Life of a cutting tool. How tool life of a cutting tool can be measured?
b) Explain the given tool signature of a single point cutting tool: $10^\circ, 8^\circ, 7^\circ, 6^\circ, 15^\circ, 30^\circ, 1/8$ (inch)

[3+5]

5. a) Explain the forces acting during oblique cutting operation by a single point cutting tool.
b) What are the desirable properties of cutting fluid?

[3+5]

6. a) What is taper? How taper can be represented?

b) A shaft 600 mm long is to be turned taper for a length of 225 mm. The amount of taper is 1:100. If the shaft diameter is 75 mm, determine the smaller diameter, half taper angle and set-over required to turn the taper.

[3+5]

7. With suitable diagram explain the process of investment casting.

[8]

8. a) State about the defects which may generate in casting process.

b) With suitable diagram explain the working principle of arc welding.

[3+5]

9. a) Explain the different types of welding flames used in gas welding.

b) State the difference between MIG welding and TIG welding.

[4+4]

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MANUFACTURING PROCESS-I*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. An example of resistance welding is - a) TIG, b) MIG, c) Submerged arc welding, d) Spot welding.
- ii. Non-Consumable electrode used in- a) GMAW, b) GTAW, c) MMAW, d) Thermit.
- iii. A welding defect is - a) crack, b) crash, c) scratch, d) swell.
- iv. In which of the following processes, material is neither added nor removed but is deformed into desired shape?- (a) Surface finishing process, (b) Metal forming process, (c) Casting, (d) Machining.
- v. Mechanical property desired for successful forging is - a) Ductility, b) Malleability, c) Hardness, d) Toughness.
- vi. The most commonly used flame in gas welding is - a) Neutral b) Oxidizing c) Carburizing d) All of the above.
- vii. Structural section such as rails, angles, I beam are made by - a) Hot extrusion, b) Hot rolling, c) Hot drawing, d) Hot spinning.
- viii. Half nut is connected with - a) Milling machine, b) Locking device, c) Jigs & Fixture, d) Thread cutting on Lathe.
- ix. Long wires are made by following process - a) Extrusion, b) Rolling, c) Piercing, d) Drawing.
- x. A moving mandrel is used in - (a) wire drawing, (b) tube drawing, (c) metal cutting, (d) casting.
- xi. Which of the following flame is harmful to steel? - (a) Neutral flame, (b) Oxidizing flame, (c) Carburizing flame, (d) Both oxidizing flame and carburizing flame.
- xii. Which of the following materials is used as the pattern material in precision investment casting? a) Metal, b) Molten wax, c) Plastics, d) Wood.
- xiii. Blanking and punching operations can be performed simultaneously on – (a) compound die, (b) combination die, (c) progressive die, (d) simple die.
- xiv. A casting defect which results in general enlargement of a casting is known as- (a) Shift, (b) sand wash, (c) swell, (d) scab.
- xv. Core print is a part of - a) core, b) pattern, c) core box, d) chaplet.

2. Fill in the blanks (any ten):

1x10

- i. Extra metal which settles down in the gutter is known as _____.
- ii. The best shape for sprue in sand-casting is _____.
- iii. Voltage required for Arc welding is _____ volts.

- iv. Oxygen cylinder used in oxy-acetylene gas welding is usually painted in _____ colour.
- v. In sand moulding, the middle part of the flask is called _____.
- vi. A _____ is a tool used for enlarging a previously drilled hole.
- vii. In blanking operation, clearance is provided on _____.
- viii. Collets, angle plates and mandrel are _____ of a lathe.
- ix. A small depression in the parent metal during arc welding is called _____.
- x. Lathe is specified by _____ of the job over carriage.
- xi. Large size bolt heads are made by _____ forging.
- xii. Model of an object which is used for making mould in sand moulding is called _____.
- xiii. Collapsible paste tubes are produced by _____.
- xiv. In _____ process, a specified shape of metal from the side or edge of a metal sheet can be removed using suitable punch and die.
- xv. _____ is a mixture of finely divided aluminium and iron oxide.

3. Answer the following questions (any ten):

1x10

- i. What is a chaplet?
- ii. Why electrodes are coated in manual arc welding?
- iii. State the relationship between lead and pitch of a thread?
- iv. Sketch three types of gas welding flames.
- v. Which sheet metal operation is used to make coins?
- vi. What is used to reduce friction during machining operation?
- vii. Define orthogonal cutting.
- viii. Pressurised gating system is used for casting of which metals and alloys?
- ix. Write down Taylor tool life equation?
- x. What is the use of filler metal in welding?
- xi. What is core print?
- xii. Which manufacturing process is used for manufacturing of Crankshaft?
- xiii. Define tool life for a single point HSS cutting tool in lathe operation.
- xiv. Which welding is used for sheet metal?
- xv. In which furnace cast iron is melted?

Group B

4. Answer the following questions (any six):

2x6

- i. What is the difference between brazing and soldering?
- ii. State the purpose of using cutting fluids in machining operations.
- iii. Name different defects of rolling process.
- iv. Give examples of single point cutting tool & multi point cutting tool.
- v. State specification of a Centre lathe.
- vi. Write two advantages of submerged arc welding.
- vii. What is Carburising flame?
- viii. Differentiate between hot working process and cold working process.
- ix. What are the components of a carriage?
- x. What is swell?

Group C

5. Answer the following question (any one):

1x6

- i. How manufacturing processes are classified? Explain with example.
- ii. Explain hydrostatic extrusion process with a simple diagram.
- iii. Briefly explain the processes - a) cutting, b) lancing, c) embossing.

5+1

6

2+2+2

6. Answer the following question (any one): 1x6
- i. Write down the favourable conditions for continuous and discontinuous chip formation? 6
 - ii. Write short notes on – a) Facing, b) Parting-off, c) Knurling. 2+2+2
 - iii. What are the Desirable properties of cutting tool materials and cutting fluid? 3 + 3
7. Answer the following question (any one): 1x6
- i. Describe the processes TIG and MIG briefly. 3+3
 - ii. Describe submerged arc welding process with sketch. 6
 - iii. Name the various types of defects that occur in sand castings. State causes and remedies in brief. 2+4
-

March 2021

MANUFACTURING PROCESSES – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any twenty): 1x20
- i) Fluxes are used while welding – (a) To increase the rate of welding, (b) To clean the joint, (c) To prevent oxidation of metal during welding, (d) All the above.
 - ii) Oxy-acetylene flame is used for welding of – (a) Monel, (b) Nickel, (c) Steel, (d) Bronze.
 - iii) The hidden defect of a weld can be detected by – (a) Tensile test, (b) Bend test, (c) Radiography, (d) All.
 - iv) Commonly used gases in TIG welding are – (a) Hydrogen & Oxygen, (b) Hydrogen & Argon, (c) Hydrogen & Helium, (d) Helium & Argon.
 - v) Consumable electrode is used in the following process – (a) TIG welding, (b) Gas welding, (c) Arc welding, (d) Resistance welding.
 - vi) Above _____ temperature, working of metals is named hot working.
 - vii) Drawing is done to _____ the diameter of wire.
 - viii) In 4-high rolling mill, bigger rolls are known as _____ rolls.
 - ix) Accurate taper and step tapers can be machined with _____ attachment.
 - x) Finishing and sizing a hole is called _____.
 - xi) Spot welding is one of the _____ welding processes.
 - xii) The revolving cutter of a milling machine is held on a _____.
 - xiii) In universal milling machine table can have the _____ arrangement about a vertical axis.
 - xiv) Mechanical property desired for successful forging is – (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness
 - xv) Seamless tubes in mass production are manufactured by – (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - xvi) In drop forging, forging is done by dropping – (a) The work piece at high velocity, (b) The hammer at high velocity, (c) The die with hammer with high velocity, (d) A weight on hammer to produce the requisite impact.
 - xvii) Blanking and punching operations can be performed simultaneously on – (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - xviii) A twist drill has _____ flutes.
 - xix) Pattern used in precision investment casting normally made of _____.

- xx) In resistance welding _____ is generated by the resistance offered to the flow of current.
- xxi) MIG welding uses _____ electrode.
- xxii) In neutral flame in Oxyacetylene gas welding, ratio of acetylene to oxygen is _____.

2. What is core print? What is core? Write the advantages of wood as pattern making material. 2+3+5
 3. How a Lathe machine can be specified? Write the functions of Lathe machine parts? 5+5
 4. Write the difference between up milling and down milling. Write the specifications of drilling machine. 5+5
 5. Write shorts on – (a) Facing, (b) Parting-off, (c) Knurling. 3+3+4
 6. Write the functions of different parts of a radial drilling machine. 10
 7. Describe hot chamber and cold chamber die-casting method. 5+5
 8. Write short notes on – (a) Sweep pattern, (b) Loose-piece pattern. 5+5
 9. Describe submerged arc welding process. 10
 10. Describe different types of milling cutter. 10
 11. Write short notes on different types of chips. 10
-

MANUFACTURING PROCESSES-I**841**

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier. Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty):

1x20

- i. Which of the following processes is mainly used for making the connecting rod?
a) Casting b) Deep drawing c) Rolling d) Forging
- ii. The hot working of metals is carried out at _____.
a) The recrystallisation temperature b) Below the recrystallisation temperature c) Above the recrystallisation temperature d) Any temperature
- iii. Which of the following is a type of extrusion process?
a) Direct b) Indirect c) Impact d) All of the mentioned
- iv. Long wires are made by _____ process.
a) Forging b) Rolling c) Piercing d) Drawing
- v. A moving mandrel is used in _____.
a) Wire drawing b) Tube drawing c) Metal cutting d) Casting
- vi. A two high rolling mill consists of two rolls which rotate _____.
a) At the same speed and in the same direction b) At the same speed but opposite direction c) At different speeds and in the same direction d) At the different speeds and in the opposite direction
- vii. In press, which of the following mechanism / system may be used for applying power to ram?
a) Rack and pinion b) Pneumatic c) Hydraulic d) All of these
- viii. In which sheet metal operation, the punch out portion of the sheet metal is the useful part?
a) Blanking b) Piercing c) Coining d) Embossing
- ix. Which of the following part is used for holding the metal sheet during blanking operation?
a) Spherical steel ball b) Roller c) Pressure pad d) Magnet
- x. Which type of chip is formed during machining of soft and ductile material?
a) Continuous chip b) Discontinuous chip c) continuous chip with built-up edge d) None of these
- xi. How many forces are acting during orthogonal cutting action?
a) One b) Two c) Three d) Four
- xii. Which type of standard taper is being provided on the shank of Lathe centers?
a) Morse taper b) Jarno taper c) Brown and Sharpe taper d) None of these
- xiii. Pitch of the lead screw of a metric Lathe is _____.
a) 4 mm b) 6 mm c) 8 mm d) 10 mm
- xiv. The size of a Lathe is specified by the _____.
a) Length between centers b) Swing diameter over the bed c) Swing diameter over the carriage d) All of these
- xv. To perform the facing operation in a Lathe, which type of feed has to be given on cutting tool?
a) Longitudinal feed b) Cross feed c) Inclined feed d) None of these
- xvi. Which specific gear is required in change gear for cutting metric thread in an English Lathe?
a) Gear of 20 teeth b) Gear of 40 teeth c) Gear of 80 teeth d) Gear of 127 teeth

- xvii. During machining, cutting fluids are used for:
a) Cool the cutting tool b) Wash away the chips c) Cool the workpiece d) All of these
- xviii. In which Lathe operation, the effective diameter of the job get increased?
a) Turning b) Facing c) Grooving d) Knurling
- xix. Which allowance is incorporated in pattern making to compensate the shrinkage of metal during solidification?
a) Machining allowance b) Distortion allowance c) Draft allowance d) Shrinkage allowance
- xx. In order to produce hollow interior of the casting _____ are used.
a) Riser pins b) Cores c) Vent tubes d) None of these
- xxi. "Master pattern" and "lost pattern" are used in _____ casting process.
a) Continuous b) Centrifugal c) Investment d) None of these
- xxii. In gas welding, how many types of flames are used for welding?
a) One b) Two c) Three d) Four
- xxiii. In which gas welding flame three distinct cones are obtained?
a) Oxidizing flame b) Carburizing flame c) Neutral flame d) None of these
- xxiv. In TIG welding, tungsten acts as _____ electrode.
a) Consumable b) Non-consumable c) Filler d) None of these
- xxv. Which welding process requires filler material?
a) Metal arc welding b) MIG welding c) TIG welding d) Resistance welding

Group B

- | | | |
|-------|--|-----|
| 2.a. | What is hot forging and cold forging? | 2+6 |
| b. | With suitable diagram explain the open die forging process. | |
| 3.a. | State the major advantages of forging process. | 2+6 |
| b. | With suitable diagram explain the indirect extrusion process. | |
| 4.a. | State the difference between orthogonal and oblique cutting. | 4+4 |
| b. | State the desirable properties of cutting tool materials. | |
| 5.a. | Define longitudinal, cross and inclined feed in respect of Lathe. | 3+5 |
| b. | With suitable diagram show the nomenclature of a single point cutting tool. | |
| 6.a. | How will you specify a Lathe? | 4+4 |
| b. | State the difference between steady rest and follower rest. | |
| 7.a. | Define odd thread and even thread. | |
| b. | Calculate the teeth number of gears and draw the gear arrangement of the change gear for cutting a screw having 7 mm pitch on a Lathe having lead screw of 4 threads per inch. | 2+6 |
| 8.a. | State the name of allowances generally being incorporated in pattern making. | 2+6 |
| b. | With suitable diagram explain the centrifugal casting process. | |
| 9.a. | State the advantages and disadvantages of welding process. | 4+4 |
| b. | State the difference between gas welding and arc welding. | |
| 10.a. | State the function of flux on a coated electrode used in arc welding. | 3+5 |
| b. | With suitable diagram explain the principle of submerged arc welding. | |

THERMAL ENGINEERING – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.

Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.

Answer five questions from Group-A & B, as directed.

1. A. Answer the following questions (any ten): 10x1
- i) Which of the following is a non-renewable resource? – (a) Coal (b) Forests (c) Water (d) Wildlife.
 - ii) Which of the following is a disadvantage of most of the renewable energy sources? – (a) Highly polluting (b) High waste disposal cost (c) Unreliable supply (d) High running cost.
 - iii) Both power and manure is provided by – (a) Nuclear plants (b) Thermal plants (c) Biogas plants (d) Hydroelectric plant.
 - iv) The amount of heat required to raise the temperature of the unit mass of gas through one degree at constant volume, is called – (a) specific heat at constant volume (b) specific heat at constant pressure (c) kilo Joule (d) none of these.
 - v) An adiabatic process is one in which – (a) no heat enters or leaves the gas (b) the temperature of the gas changes (c) the change in internal energy is equal to the mechanical work done (d) all of the above.
 - vi) Work done in a free expansion process is – (a) zero (b) minimum (c) maximum (d) positive.
 - vii) The ratio of specific heat at constant pressure (C_p) and specific heat at constant volume (C_v) is – (a) equal to one (b) less than one (c) greater than one (d) none of these.
 - viii) The value of gas constant (R) in S.I. units is – (a) 0.287 J/kgK (b) 2.87 J/kgK (c) 28.7 J/kgK (d) 287 J/kg.
 - ix) With usual notation $\Delta x/kA$ is called – (a) Thermal resistance (b) Thermal conductance (c) Heat flux density (d) Thermal loading.
 - x) Pressure for Water at critical point – (a) 221.2 bar (b) 221.2 N/sq.m (c) 225.65 kgf/cm² (d) None of the above.
 - xi) The sum of internal energy (U) and the product of pressure and volume ($p.v$) is known as – (a) workdone (b) entropy (c) enthalpy (d) none of these.
 - xii) The boiling point of steam depends on – (a) Temperature (b) pressure (c) Enthalpy (d) all of the above.

B. State whether the following statements are True or False (any ten):

10x1

- i) In the first law of thermodynamics, the total energy of the system remains constant.
- ii) The kinetic energy of molecules of a gas becomes zero at absolute zero temperature.
- iii) Entropy is a point function.
- iv) The solid fuels have higher calorific value than liquid fuels.
- v) At the time of condensation heat released by steam is sensible heat.
- vi) Steam is a non homogeneous compound.
- vii) 2nd law of thermodynamics indicates the direction of heat transfer.
- viii) A black body reflects all the radiation energy.
- ix) PMM of the second kind is always possible to construct.
- x) Work done is a path function.
- xi) At triple point only steam exists.

Group-A

Answer any three questions.

- 2. Classify energy sources. Discuss disadvantages of Renewable energy. Write short notes on Geothermal Energy, Fossil Fuel, Green Energy. 2+2+6
- 3. Discuss Zeroth law of Thermodynamics. Derive Steady flow energy equation. Write down short notes on Intensive and Extensive properties, quasi static process. 2+4+4
- 4. A non flow Quasi static process (Reversible) occurs for which $p = (-3V+16)$ bar, where V is volume in m^3 . What is the work done if V changes from 2 to 6 m^3 . Derive $PV^n = \text{Const.}$ (with usual notation) 4+6
- 5. Define heat engine, Refrigerator, Heat pump. Prove Reversible engines working between two fixed temperature always be more efficient than any other engine working between same two fixed temperatures. 6+4
- 6. 0.45 kg of Air at 179°C expands adiabatically and reversibly to three times its original volume and during the process there is a fall in temperature to 15.5°C . The work done during the process is 53 kJ. Calculate C_p and C_v . Derive the expression of Enthalpy for constant pressure non flow process. 7+3

Group-B

Answer any two questions.

- 7. Define Dryness fraction, Triple point, Critical point, Super heated steam. Discuss phase transformation with diagram. (4x2)+2
- 8. Find the enthalpy, Internal Energy, and Entropy of 1 kg of steam at a pressure of 10 bar when steam is (a) Dry saturated, (b) When steam is .75 Dry, (c) When steam is at 250°C . Use steam table. 10
- 9. Discuss Fourier's Law of Conduction. Derive an expression of conduction through a flat composite wall of several homogeneous layers. 3+7
- 10. Discuss cross flow heat exchanger. Define Grey body, Black body, Absorptivity, Stefan Boltzman law. 2+(4x2)

THERMAL ENGINEERING - I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.

Answer any five questions from Group-A & B, as directed.

1. Answer the following questions (any twenty):

20x1

Choose the correct alternatives from the following:

- i) A closed system receives 60 kJ of heat but its internal energy decreases by 30 kJ. Then the work done by the system is – (a) -90 kJ, (b) -30 kJ, (c) 30 kJ, (d) 90 kJ.
- ii) The internal energy (u) of a dry steam is given by, [p is in bar, v_g is in m^3/kg , u is in kJ/kg] – (a) $u = h_g + 100 p v_g$ (b) $u = h_g - 100 p v_g$, (c) $u = h_g / 100 p v_g$ (d) $u = 100 p v_g$.
- iii) Which of the following is an intensive property of a thermodynamic system? – (a) volume, (b) temperature, (c) energy, (d) mass.
- iv) A composite wall of a furnace has 3 layers of equal thickness having thermal conductivities in the ratio of 1:2:4. What will be the temperature drop ratio across the three respective layer? – (a) 1:2:4, (b) 4:2:1, (c) 1:1:1, (d) $\log 4: \log 2: \log 1$.
- v) Entropy may be expressed as a function of – (a) pressure and temperature, (b) temperature and volume, (c) heat and work, (d) both (a) & (b).
- vi) The value of universal gas constant (R_u) is – (a) 83.14 J/kgK, (b) 8.314 J/kgK, (c) 831.4 J/kgK, (d) 8314 J/kgK.
- vii) In 100 kg wet saturated steam, mass of the water particle is 10 kg. The dryness fraction of the steam will be – (a) 0.1, (b) 1, (c) 0.85, (d) 0.9.
- viii) The change in internal energy is zero during – (a) adiabatic process, (b) polytropic process, (c) isothermal process, (d) throttling.
- ix) When gas is heated at constant pressure – (a) raises the temperature of the gas, (b) increases the internal energy of the gas, (c) does some external work during expansion, (d) both (b) & (c), (e) both (a) & (b).
- x) The general law of expansion or compression is $PV^n = \text{constant}$. The process is said to be hyperbolic if n is equal to – (a) 0, (b) γ , (c) 1, (d) ∞ .
- xi) If a heat engine gives an output of 3 kW when the input is 10000 J/sec, then the thermal efficiency of the engine will be _____.
- xii) The specific volume of water _____ when heated from 0°C . (decreases/ /increases/ first decreases, then increases / first increases, then decreases)

- xiii) The emissivity is highest for – (a) Rubber, (b) Paper, (c) iron oxide, (d) Carbon.
- xiv) The highest value of thermal conductivity is expected for – (a) Water, (b) Solid ice, (c) Melting ice, (d) Steam.

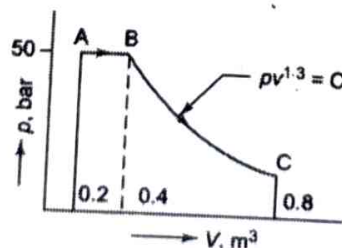
State whether the following statements are True or False:

- xv) The perpetual motion machine of first kind is possible.
- xvi) Zeroth law of thermodynamics deals with temperature.
- xvii) Ratio of two specific heats of a perfect gas is always varies with temperature.
- xviii) In isothermal process change in internal energy is zero.
- xix) Efficiency of any heat engine cycle is always 100%.
- xx) Entropy of universe is increasing.
- xxi) Quasistatic process is a reversible process.
- xxii) Heat is low grade energy.

Group-A

Answer any three questions.

2. a) Write down the advantages and disadvantages of Geothermal Energy.
b) Write short notes on: - Solar water heater, Biogas plant. 4+6
3. a) Differentiate between extensive and intensive properties with example.
b) Prove that heat transfer is a path function.
c) Show that the work done for an isothermal process from the state 1 to state 2 is given by $W_{1-2} = p_1 v_1 (\ln p_1 - \ln p_2)$, where each term has its usual meaning. 3+3+4
4. a) What is steady flow process? Write the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms of it.
b) A gaseous system undergoes three quasi-static processes in sequence. The gas initially at 5 bar and 0.01 m^3 is expanded at constant pressure. It is then further expanded according to the relationship $p v^2 = \text{constant}$ to 2 bar and 0.025 m^3 . The gas is then returned to its initial state during which $p v = \text{constant}$. Sketch the cycle in the P-V diagram. Calculate the work done in each case. (1+4)+(1+4)
5. a) Determine the total work done by a gas system following an expansion process as shown in Figure.



- b) A mass of 8 kg gas expands within a flexible container so that the p-v relationship is of the form $p v^{1.2} = \text{constant}$. The initial pressure is 1000 kPa and the initial volume is 1 m^3 . The final pressure is 5 kPa. If the specific internal energy of the gas decreases by 40 KJ/Kg, find the heat transfer in magnitude and direction. 5+5
6. a) Prove that for an ideal gas, change in entropy, $S_2 - S_1 = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$.
b) An inventor claims to have heat engine which is capable of developing 9 kW while working between the temperature limits of 30°C and -40°C . It receives only 1000 kJ/min of heat. Discuss the possibility of the claim. 5+5

7. a) Establish the relation: $(C.O.P.)_{HP} = (C.O.P.)_R + 1$
 b) A mass of 50 kg/min enters the control volume of a steady flow system at 2 bar and 100°C and an elevation of 100 m above the datum. The same mass leaves the control volume at 150 m elevation with a pressure of 10 bar and temperature of 300°C. The entrance and exit velocities are 2400 m/min and 1200 m/min respectively. During the process, 50,000 kJ/hr of heat are transferred to the control volume and the rise in the enthalpy is 8 kJ/kg. Calculate the power developed. 3+7
8. a) State Kelvin-Planck and Celsius statements related to 2nd law of thermodynamics.
 b) 10 kg of air is compressed adiabatically from 1 bar pressure and 18°C to 7 bar and then cooled at constant pressure to original temperature. The air is then expanded adiabatically to the original pressure. [Take $\gamma = 1.4$ and $C_p = 1.005 \text{ kJ/kgK}$]. Find – (i) The temperature after compression, (ii) The final temperature and (iii) Heat extracted during constant pressure cooling. 4+6

Group-B

Answer any two questions.

9. a) Describe the working principle of Throttling calorimeter with neat sketch.
 b) A rigid closed tank of volume 3 m³ contains 5 kg of wet steam at a pressure of 200 kPa. The tank is heated until the steam becomes dry saturated. Determine the final pressure and the heat transfer to the tank. 4+6
10. a) Define dryness fraction of steam. What do you mean by degree of superheat?
 b) Write down the advantages of using Superheated steam.
 c) In a throttling calorimeter, the steam is admitted at a pressure of 10 bar. If it is discharged at atmospheric pressure and 110°C after throttling, determine the dryness fraction of steam. Assume specific heat as steam as 2.2 kJ/kg K. 4+3+3
11. a) State Fourier's law of heat conduction.
 b) A composite wall consisting of three different layers in perfect thermal contact is shown in sketch below. The outer surfaces to the extreme left and right are kept at temperatures 400°C and 50°C respectively. The thickness L_i for $i = 1, 2, 3$ of each layer are given as 5cm, 10cm and 15 cm and the thermal conductivity k_i for $i = 1, 2, 3$ of each layer are given as 20W/m°C, 50W/m°C and 100W/m°C respectively. Determine the heat transfer rate per square meter across this composite layer by assuming one dimensional heat flow and using the thermal resistance concept. 3+7
12. a) Distinguish between – (i) Black body and gray body, (ii) Absorptivity and emissivity of a surface.
 b) Deduce the formula of overall heat transfer co-efficient for a flat composite wall, taking into consideration the convective heat transfer. 4+6

March 2021

THERMAL ENGINEERING – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any twenty):

1×20

State True or False:

- i) During throttling process enthalpy remains constant.
- ii) H-S diagram for water steam is called Mollier diagram.
- iii) Superheated steam behaves like a perfect gas.
- iv) During heat transfer by radiation no medium is required.
- v) Intensive properties per unit mass are extensive properties.
- vi) A process with identical end states is called a cycle.
- vii) PMM-II violates the first law of thermodynamics.

Choose correct answer from the given alternatives:

- viii) Which of the following relation is correct? -
a) $C_p/C_v=R$ b) $C_p+C_v=R$ c) $C_p/C_v=\gamma$ d) $C_p-C_v=\gamma$
- ix) For same temperature limit which relation will be correct? -
a) $1 - \text{COP} = \eta_{\text{HE}}$ b) $1 + \text{COP} = \eta_{\text{HE}}$
c) $\text{COP} - \eta_{\text{HE}} = 1$ d) None of these
- x) The work transfer for isochoric process will be -
a) $P_1 V_1 \ln P_1/P_2$ b) $(P_1 V_1 - P_2 V_2)/(N-1)$
c) Zero d) None of these
- xi) For which of the following processes Heat Transfer will be zero? -
a) Adiabatic b) Isochoric c) Isobaric d) None of these
- xii) Dryness fraction of wet saturated steam will be -
a) >1 b) <1 c) $=1$ d) 0
- xiii) Which of the following relation is true? -
a) $1 = \alpha + \beta + \gamma$ b) $\alpha + \beta - 2\gamma = 2$ c) $\alpha - \beta - \gamma = 0$ d) $\alpha\beta = \gamma$
- xiv) Which of the following is renewable source of energy? -
a) Water b) Coal c) Natural Gas d) None of these
- xv) Turbine are used in following power plant -
a) Solar b) Thermal c) Nuclear d) Both b & c
- xvi) A cup of tea place on a table is an example of -
a) Open system b) Closed System
c) Isolated system d) None of these
- xvii) Which will be is unit of Power given below? -
a) KJ/Kg b) KJ/hr c) Cal/Kg d) KJ-hr

Fill in the blanks:

- xviii) The value of universal gas constant is _____.
xix) The work done in free expansion process is _____.
xx) The locus of saturated liquid line and saturated vapor line meets at _____.
xxi) The latent heat of steam at atmospheric pressure is _____.
xxii) The thermal conductivity of solid metals _____ with rise in temperature.
xxiii) Summation of internal energy and _____ is called enthalpy.
xxiv) Atmospheric pressure is equal to _____ bar.
xxv) Kelvin-Planks law deals with _____.
2. Define heat engine, Refrigerator, Heat pump. Prove Reversible engines working between two fixed temperature always be more efficient than any other engine working between same two fixed temperatures. 6+4
3. 0.45 kg of Air at 179°C expands adiabatically and reversibly to three times its original volume and during the process there is a fall in temperature to 15.5 °C. The work done during the process is 53 kJ. Calculate C_p and C_v . Derive the expression of Enthalpy for constant pressure non flow process. 7+3
4. Define Dryness fraction, Triple point, Critical point, Super heated steam. Discuss phase transformation with diagram. (4x2)+2
5. Find the enthalpy, Internal Energy, and Entropy of 1 kg of steam at a pressure of 10 bar when steam is (a) Dry saturated, (b) When steam is .75 Dry, (c) When steam is at 250°C .Use steam table. 10
6. Classify energy sources. Discuss disadvantages of Renewable energy. Write short notes on Geothermal Energy, Fossil Fuel, Green Energy. 2+2+6
7. A non flow Quasi static process (Reversible) occurs for which $p = (-3V + 16)$ bar, where V is volume in m^3 . What is the work done if V changes from 2 to 6 m^3 . Derive $PV^n = \text{Const.}$ (with usual notation) 4+6
8. Discuss Fourier's Law of Conduction. Derive an expression of conduction through a flat composite wall of several homogeneous layers. 3+7
9. Discuss cross flow heat exchanger. Define Grey body, Black body, Absorptivity, Stefan Boltzman law. 2+(4x2)
10. Discuss Zeroth law of Thermodynamics. Derive Steady flow energy equation. Write down short notes on Intensive and Extensive properties, quasi static process. 2+4+4
-

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THERMAL ENGINEERING-I

Time Allowed: 3 Hours

Full Marks: 60

Answer the following questions from Group-A, B & C as directed.

Group A

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. Heat pump works on the principle of - a) Kelvin plank Statement, ~~b) Clausius statement~~, c) Zeroth law of thermodynamics, d) energy equation of flow process.
- ii. In steady flow process - ~~a) the mass flow rate is constant~~, b) the heat transfer rate is constant, c) the work transfer rate is constant, d) all of these.
- iii. Superheated vapour behaves - a) Exactly as gas, b) As steam, c) As ordinary vapour, d) Approximately as a gas.
- iv. In an economiser - a) Feed water is heated by the heat extracted from flue gas, b) Primary air is heated by the heat extracted from the flue gas, c) Suspended solids are separated from the flue gas using electrostatic forces, d) Feed water is pumped to the de-aerator.
- v. Dryness fraction of a steam containing 20% of water vapour by mass is - a) 0.2, b) 80, c) 0.8, d) 20.
- vi. Heat and work are - (a) intensive properties, (b) extensive properties, (c) point functions, ~~(d) path functions~~.
- vii. Throttling of steam during dryness fraction determination by throttling calorimeter is essentially - a) isenthalpic process, b) isentropic process, c) isochoric process, d) isobaric process.
- viii. A series of operations, which take place in a certain order and restore the initial condition is known as- ~~a) reversible cycle~~, b) irreversible cycle, c) thermodynamic cycle, d) None of these.
- ix. The value of C_p/C_v for air is - ~~a) 1~~, b) 1.4, c) 1.8, d) 2.3.
- x. Mollier chart is also known as - a) T-S diagram b) p-T diagram c) p-V diagram d) h-S diagram
- xi. Which of the following statements is wrong? - a) Mechanical draught reduces the height of chimney., b) Natural draught reduces the fuel consumption., c) A balanced draught is a combination of induced and forced draught., d) All of the statements.
- xii. For a reversible adiabatic process, the change in entropy is - a) minimum, b) maximum, c) unity, ~~d) zero~~.
- xiii. According to Clausius statement of the second law of thermodynamics, the coefficient of performance of a refrigerator is never- (a) infinite, (b) greater than unity, (c) unity, ~~(d) less than unity~~.
- xiv. A device used in boiler to control the flow of steam from the boiler to the main pipe and shut off the steam completely when required, is known as - a) blow off cock, b) fusible plug, ~~c) stop valve~~, d) economiser.
- xv. Water tubes in a Babcock and Wilcox boiler are - ~~a) Horizontal~~, b) Vertical, c) Inclined, d) inclined.

2. Fill in the blanks (any ten):

1x10

- i. PMM2 violates _____ law.
- ii. Work is a _____ function.
- iii. Quantity of steam generated by a boiler at full load is called _____ of the boiler.
- iv. COP of heat pump is _____ than COP of refrigerator.
- v. First law of thermodynamics refers to the conservation of _____.
- vi. Thermal efficiency of a Carnot engine working between temperature levels T_1 and T_2 ($T_1 > T_2$) is given by _____.
- vii. Change of internal energy is zero during _____ process.
- viii. The unit of pressure in S.I. unit is _____.
- ix. One example of vertical fire tube boiler is _____.
- x. In La-mont boiler there is _____ circulation of water and steam.
- xi. For reversible adiabatic process _____ remains constant.
- xii. At critical point, the enthalpy of vaporisation i.e h_{fg} , is equal to _____.
- xiii. Total heat before throttling is _____ to total heat after throttling.
- xiv. As per Charles law, volume of given mass of a perfect gas varies _____ with its absolute temperature, when the absolute pressure remains constant.
- xv. Absolute zero temperature is _____.

3. Answer the following questions (any ten):

1x10

- i. State Boyle's law.
- ii. What is meant by quasi-static process in thermodynamics?
- iii. Differentiate universal gas constant and characteristic gas constant.
- iv. Give two examples of boiler accessories.
- v. Define Evaporation Ratio of a boiler.
- vi. Give one example of an open system.
- vii. Define Dryness fraction.
- viii. State Avogadro's law.
- ix. What is an isolated system?
- x. Write the value of characteristic gas constant?
- xi. What is superheated steam?
- xii. Under what circumstances, steam can be treated as an ideal gas?
- xiii. Why work is treated as high grade energy?
- xiv. Define a pure substance?
- xv. State the use of a calorimeter.

Group B

4. Answer the following questions (any six):

2x6

- i. Calculate the work done by a gas in an isothermal process from A to B. $P_A = 1 \text{ Pa}$, $V_A = 3 \text{ m}^3$ and $P_B = 3 \text{ Pa}$.
- ii. How does fusible plug work in fire tube boiler?
- iii. Explain in brief the concept of flow work.
- iv. What do you mean by entropy of wet steam?
- v. State Kelvin Planck statement.
- vi. Briefly state the function of an air-preheater.
- vii. Show different thermodynamic processes on P-V and T-S diagram.
- viii. Define a perfect gas.
- ix. Why circulating pump is required for high pressure boilers?
- x. What is latent heat of vaporisation?

Group C

5. Answer the following question (any one): 1x6
- i. A Carnot heat engine receives heat from a reservoir at 900°C at a rate of 800 kJ/min and reject the waste heat to the ambient air at 27°C . The entire work output of the engine is used to drive a refrigerator that removes heat from the refrigerated space at -5°C and transfers it to the same ambient air at 27°C . What will be the maximum rate of heat removal (in kJ/min) from the refrigerated space? 6
 - ii. Define the following with examples: a) Open system, b) Closed system, c) Isolated system. 2+2+2
 - iii. State First Law of Thermodynamics. What are the limitations of this law? Give an example of application of the law. 2+3+1
6. Answer the following question (any one): 1x6
- i. Show that $C_p - C_v = R$. 6
 - ii. Prove that during adiabatic process, work done is given by $W = (P_1 V_1 / \gamma - 1) \{1 - P_2 V_2 / P_1 V_1\}$. 6
 - iii. Steam expands isentropically in a nozzle from 1 MPa , 250°C to 10 kPa . The steam flow rate is 1 kg/s . Neglecting the K.E of steam at inlet to the nozzle, find the velocity of steam at exit from the nozzle and the exit area of the nozzle ? 6
7. Answer the following question (any one): 1x6
- i. In a separating and throttling calorimeter the total quantity of steam passed was 40 kg and 2.2 kg of water was collected from separator. Steam pressure before throttling was 1.5 MPa and temperature and pressure after throttling are 120°C and 105 kPa . Determine the dryness fraction of steam before entering to calorimeter. Specific heat of superheated steam may be considered as 2.09 kJ/kg.K . 6
 - ii. Draw a simple neat sketch of Cochran boiler, Label all its components and briefly mention some common features of the boiler. 6
 - iii. What are boiler mountings and accessories? What is the difference between them? Give 3 examples each of boiler mountings and accessories. 2+2+2

THERMAL ENGINEERING-I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): 1x20
 - a. The value of $n = 1$ in the polytropic process indicates it to be – (a) adiabatic process (b) isobaric process (c) isothermal process (d) isentropic process
 - b. Measurement of temperature is based on _____ law of thermodynamics – (a) zeroth (b) first (c) second (d) third
 - c. An open system is one in which _____. (a) Mass does not cross boundaries of the system, though energy may do so (b) Neither mass nor energy crosses the boundaries of the system (c) Both energy and mass cross the boundaries of the system (d) Mass crosses the boundary but not the energy
 - d. The first law of Thermodynamics relates to – (a) Conservation of mass (b) Conservation of momentum (c) Conservation of internal energy (d) Conservation of energy
 - e. Properties of substances like pressure, temperature and density, in thermodynamic coordinates are – (a) path functions (b) point functions (c) cyclic functions (d) real functions
 - f. “No process is possible whose sole result is the transfer of heat from a colder object to a hotter object” this statement is related to the – (a) zeroth law of thermodynamics (b) first law of thermodynamics (c) Kelvin Planck statement (d) Clausius statement
 - g. A machine which can supply mechanical work continuously without consumption of any energy is called as – (a) PMM1 (b) PMM2 (c) PMM3 (d) all the above
 - h. Boyle’s law relates to – (a) Constant volume (b) Constant pressure (c) Constant temperature (d) none of the above
 - i. For a perfect gas the specific heat at constant pressure C_p and that at constant volume C_v are related as – (a) $C_p = C_v$ (b) $C_p = C_v + R$ (c) $C_p = C_v - R$ (d) $C_p = C_v + \log R$
 - j. The statement that molecular weights of all gases occupy the same volume is known as – (a) Dalton’s law (b) Joule’s law (c) Boyle’s law (d) Avogadro’s hypothesis
 - k. The conditions at which a real gas behaves as an ideal gas – (a) high pressure (b) low temperature (c) low pressure (d) at any pressure
 - l. When a gas is compressed adiabatically, its temperature – (a) remains constant (b) increases (c) decreases (d) none of the above
 - m. The point that connects the saturated liquid line and saturated vapour line is called – (a) saturated point (b) critical point (c) triple point (d) superheated point

- n. Dryness fraction of dry saturated steam on heating – (a) increases (b) decreases (c) remain same (d) cannot predict
- o. Latent heat of vapourisation of steam – (a) increases with the increase of pressure (b) decreases with the increase of pressure (c) remains constant at all pressures (d) latent heat doesn't depend upon pressure
- p. Measure the dryness fraction of a very wet steam with the help of – (a) separating calorimeter (b) throttling calorimeter (c) combined separating and throttling calorimeter (d) bomb calorimeter
- q. The critical pressure of water is – (a) 7.38 MPa (b) 22.12 MPa (c) 100 MPa (d) 5.04 MPa
- r. The isentropic process on the Mollier chart is represented by – (a) horizontal line (b) vertical line (c) inclined line (d) curved line
- s. Equivalent evaporation is the amount of water evaporated in a boiler from and at – (a) 0°C (b) 100°C (c) saturation temperature at given pressure (d) room temperature
- t. Cochran boiler is a – (a) horizontal fire tube boiler (b) horizontal water tube boiler (c) vertical fire tube boiler (d) vertical water tube boiler
- u. A fusible plug is fitted in small boilers in order to – (a) control steam flow (b) extinguish fire if water level in boiler falls below alarming limit (c) remove ash (d) all the above <https://www.wbsctonline.com>
- v. The draught in boiler may be produced by – (a) mechanical fan (b) chimney (c) a steam jet (d) all of these
- w. _____ is NOT a boiler mounting. – (a) Blow off cock (b) Feed check valve (c) Economiser (d) Safety valve
- x. The water tubes in a Babcock and Wilcox boiler are – (a) horizontal (b) vertical (c) curved (d) inclined
- y. The maximum heat loss in a boiler occurs due to – (a) moisture in fuel (b) dry flue gases (c) steam formation (d) unburnt carbon

Group B

2.
 - a) A cylinder contains 0.45 m^3 of a gas at $1 \times 10^5\text{ N/m}^2$ and 80°C . The gas is compressed to a volume of 0.13 m^3 , the final pressure being $5 \times 10^5\text{ N/m}^2$. Determine: (i) The mass of gas; (ii) The value of index 'n' for compression; (iii) The increase in internal energy of the gas; (iv) The heat received or rejected by the gas during compression. Take $\gamma = 1.4$, $R = 294.2\text{ J/kg}^{\circ}\text{C}$.
 - b) Make a comparison of ideal gas with a real gas. 6+2
3.
 - a) Draw a Mollier diagram showing different lines.
 - b) The following observations were recorded in an experiment with a combined separating and throttling calorimeter:
 Pressure in the steam main: 15 bar
 Mass of water drained from the separator: 0.55 kg
 Mass of steam condensed after passing through the throttle valve: 4.20 kg
 Pressure and temperature after throttling: 1 bar, 120°C
 Using steam tables, evaluate the dryness fraction of the steam in the main, and state with reasons, whether the throttling calorimeter alone could have been used for this test. 2+6
4.
 - a) State Clausius statement of second law of thermodynamics. With the help of block diagrams, show the COP of refrigerator and COP heat pump.
 - b) A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C , enthalpy 2785 kJ/kg, velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: Pressure 20 kPa, enthalpy 2512 kJ/kg, velocity 100 m/s, and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s, calculate the power output of the turbine in kW. 3+5

5. a) Explain the steam generation process with the help of p-v and T-s diagram.
b) Define wet steam, dry saturated steam and superheated steam. 5+3
 6. a) What do you mean by system? Classify systems and explain each system.
b) What do you mean by properties of a system? Define intensive properties and extensive properties with proper examples. 4+4
 7. a) Prove that adiabatic process follows the law $PV^\gamma = \text{const.}$
b) A cylinder contains 0.02 m^3 of air at 1.66 MPa and 50°C . The air expands isothermally to 104 KPa . Determine: i) Work transfer, ii) Heat transfer, iii) Final volume. 4+4
 8. a) List the different sources of heat losses in a boiler.
b) The following readings were taken during the test on a boiler for a period of 1 hour: steam generated 5000 kg ; coal burnt 700 kg ; CV of coal 31402 kJ/kg and quality of steam 0.92 . If the boiler pressure is 1.2 MPa and feed water temperature is 45°C , calculate the boiler equivalent evaporation and thermal efficiency. [using Steam Table] 2+6
- 1.2×10
9. a) Explain the construction and working principle of Lamont boiler with a neat sketch.
b) Distinguish between fire tube boiler and water tube boiler. 5+3
 10. a) Derive the expression, $h = 353H \left[\frac{1}{T_a} - \left(\frac{m+1}{m} \right) \frac{1}{T_f} \right]$ of natural draught in mm of water column.
(All the symbols have their usual meanings)
b) List at least three mountings and three accessories used in boiler. 5+3

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318/5(N)

DECEMBER 2024
THERMAL ENGINEERING-I

3247848

Time Allowed: 3 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and

1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

- i. Cochran boiler is a
(a) horizontal fire-tube boiler, (b) horizontal water-tube boiler,
(c) vertical water-tube boiler, (d) vertical fire tube boiler.
- ii. With increase of pressure, latent heat of evaporation
(a) increases, (b) decreases, (c) remain constant (d) may increases or decreases
- iii. Superheated vapour behaves
(a) Exactly as gas, (b) as steam, (c) as ordinary vapour, (d) approximately as gas.
- iv. Measurement of temperature is based on _____ law of thermodynamics.
(a) zeroth, (b) first, (c) second, (d) third.

- v. Boyle's law relates to
(a) constant volume, (b) constant pressure, (c) constant temperature, (d) none of these.
- vi. Water tube boilers are those in which
(a) flue gases pass through tubes and water around it, (b) water passes through the tubes and flue gases around it, (c) work is done during adiabatic expansion, (d) change in enthalpy.
- vii. The enthalpy of a substance is defined as
(a) $H = U - PV$, (b) $H = U + PV$, (c) $H = -U + PV$, (d) $H = -U - PV$.
- viii. When water is heated with the rise of temperature, it consumes
(a) latent heat, (b) sensible heat, (c) both (a) and (b) (d) none of these.
- ix. Heat is
(a) intensive properties, (b) extensive properties, (c) both intensive and extensive properties, (d) can't say



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- x. The temperature at which the Celsius and Fahrenheit scales read the same is
(a) 60° (b) -60° (c) -40° (d) 40°
- xi. The maximum heat loss in a boiler occurs due to
(a) moisture in fuel, (b) dry flue gases, (c) steam formation, (d) unburnt carbon.
- xii. Dryness fraction(x) of superheated steam is
(a) equal to 0, (b) greater than 1, (c) less than 1, (d) equal to 1.
- xiii. The function of Blow down valve of a boiler is
(a) to remove sediments (b) to build sediments (c) to remove flue gas (d) to remove ash
- xiv. The natural draught in boiler is produced by
(a) steam jet (b) centrifugal fan (c) chimney (d) both (a) and (b)
- xv. An economiser is installed in a boiler primarily to -
(a) superheat the steam (b) reduce fuel consumption
(c) increase steam pressure (d) all of these

GROUP-B

2. a) State First law of thermodynamics. What are the limitations of this law?
b) Deduce the steady flow energy equation. [3+5]
- ***
3. a) Define with brief explanation: PMM-I and PMM-II
b) Define the following with example: (i) system, surroundings and boundary, (ii) intensive and extensive properties [4+4]
4. a) State Boyle's law and Charle's laws.
b) Show that $C_p - C_v = R$. [3+5]
5. a) What is universal gas constant?
b) 0.2 m^3 of an ideal gas at a pressure of 2 MPa and 600 K is expanded isothermally to 5 times the initial volume. It is then cooled to 300 K at constant volume and then compressed back polytropically to its initial state. Determine the net work done during the cycle. [2+6]
6. a) Deduce the expression for work done during adiabatic process.
b) Write shortly on (i) Dryness fraction of steam (ii) Air preheater of boiler [4+(2+2)]



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7. a) What is meant by critical point? Draw the outline of Mollier chart and show critical point on it.
b) How much heat is needed to convert 5kg of water at 45°C into steam at pressure of 4 bar and 80 per cent dry? Take specific heat of water as 4.187 kJ/kg-K .
[(2+2)+4]
8. a) Differentiate between fire tube boiler and water tube boiler.
b) What is feed water? What are the necessities of boiler feed water treatment?
c) Define equivalent evaporation.
[3+(1+2)+2]
9. a) Explain with neat sketch the construction and working principle of Cochran boiler.
b) A boiler uses 18 kg air per kg of fuel. The mean temperature of chimney gases is 315°C and that of outside air 27°C . Determine the minimum height of chimney required to produce a draught of 20 m.
[5+3]



THERMAL ENGINEERING - I

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

- ② i) Which of the following is the extensive property of a thermodynamic system?
 a) pressure ☒ b) volume c) temperature d) density
- ① ii) In thermodynamics, heat and work are -
 a) point functions ☒ b) path functions c) both point and path functions d) non of these
- iii) Which of the following represent an open system?
 a) an air-tight container ☒ b) Pressure cooker c) a beaker full of water d) thermos flask
- iv) The work done is minimum in the following gas process.
 a) isobaric b) isochoric c) isothermal d) adiabatic
- ① v) Which of the following law of thermodynamics is used to understand the concept of energy conservation?
 a) zeroth law ☒ b) first law c) second law d) none of these
- vi) An ideal gas is one which obeys the law -
 a) $pV=RT$ b) $p = \frac{vR}{T}$ c) $pV = \frac{R}{T}$ d) none of these
- vii) A perfect gas at 300 K is heated at constant pressure till its volume is double. The final temperature is -
 a) 150 K b) 450 K c) 600 K d) No change of Temperature
- viii) At critical point, the enthalpy of vaporisation is -
 a) dependent on temperature only b) maximum c) minimum ☒ d) zero
- ix) The dryness fraction (x) of fully wet steam is -
☒ a) $x = 0$ b) $x = 1$ c) $0 < x < 1$ d) $x > 1$
- x) If the enthalpy of saturated water and dry saturated steam at certain conditions are 1185 kJ/kg and 2790 kJ/kg respectively, then the enthalpy of wet steam of dryness fraction 0.5 is -
 a) 1185 kJ/kg b) 3975 kJ/kg c) 2580 kJ/kg d) 2790 kJ/kg
- ① xi) In a fire-tube boiler, the water is in -
 a) tube ☒ b) shell c) all of these d) non of these
- ① xii) The natural draught in boiler is produced by
 a) steam jet b) centrifugal fan ☒ c) chimney d) both (a) and (b)
- ① xiii) An economiser is installed in a boiler primarily to -
 a) superheat the steam b) reduce fuel consumption c) increase steam pressure d) all of these

- ① xiv) The amount of water evaporated 'from and at 100 °C' into dry saturated steam at standard atmospheric pressure is called-
- a) evaporative capacity b) generation factor c) boiler horse power d) equivalent evaporation
- ① xv) The safety device used to protect the boiler when the water level falls below the minimum level in the boiler is-
- a) fusible plug b) safety valve c) blow-off cock d) water level indicator

GROUP-B

- ② 2. a) State first law of thermodynamics and specify its applications.
- b) A nozzle is a device for increasing the velocity of a steadily flowing stream. At inlet to a certain nozzle, the fluid parameters are: Enthalpy is 2850 kJ/kg; velocity is 50 m/s; area is 0.1 m² and specific volume is 0.18 m³/kg. At the discharge end, the Enthalpy is 2650 kJ/kg and the specific volume is 0.49 m³/kg. The nozzle is horizontal and there is negligible heat loss from it. Find the velocity of fluid at exit from the nozzle, mass flow rate and the exit area of the nozzle. [3+5]
3. a) What is thermodynamic system? Classify it and briefly describe them with examples.
- b) What is perpetual motion machine of 2nd kind (PMM2)? Write the efficiency of PMM2. Why such a machine cannot be constructed in actual practice? [(1+3) + (2+1+1)]
- ① 4. a) State the differences between real gas and ideal gas. Under which condition does a real gas behave most like an ideal gas? State the condition at which steam behaves almost like a gas.
- b) Show that $C_p - C_v = R$ [(2+1+1) + 4]
5. a) Derive an expression for the work done during an isothermal process of an ideal gas.
- b) A quantity of gas has a volume of 0.14 m³, pressure 1.5 bar and temperature 100°C. If the gas is compressed at constant pressure, until its volume becomes 0.112 m³, determine (i) the temperature at the end of compression and (ii) work done in the process. [4 + (2+2)]
- ② 6. a) List out the advantages of superheated steam.
- b) Determine the quantity of heat required to convert 1 kg of water at 30 °C to steam at a pressure of 6 bar under the following conditions:
- when the steam is dry saturated;
 - when the steam is wet having a dryness fraction 0.85; and
 - when the steam is superheated to 250°C, Take the specific heat of water as 4.2 kJ/kg K and that of the superheated steam as 2.25 kJ/kg K. [2+6]
- ② 7. a) Explain the terms 'critical point' with respect to steam generation.
- b) Write short notes on 'Mollier diagram'.
- ② c) A boiler evaporates 3.6 kg of water per kg of coal into dry saturated steam at 10 bar. The feed water temperature is 32° C. Find the equivalent evaporation from and at 100° C. [2 + 3 + 3]

⑥ 8. a) Classify boilers based on (i) relative position of hot gas and water, (ii) method of circulation of water.
Give example of each.

b) Describe with a neat sketch the working principle of Babcock & Wilcox boiler. [3 + 5]

9. a) State the functions of safety valve and blow-off cock employed in boiler.

b) State the advantages of artificial draught system over natural draught system.

c) The following data pertain to a steam power plant:

Height of chimney = 30 m ;

Draught produced 16.5 mm of water gauge ;

Temperature of flue gas = 360°C ;.

Temperature of boiler house = 28°C ;

Determine the quantity of air used per kg of fuel burnt in the boiler. [2 + 2 + 4]

THERMAL ENGINEERING -I

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2×10=20

- (i) First law of thermodynamics says about
 (a) Quality of energy (b) quantity of energy (c) both of (a) & (b) (d) none of these
- (ii) Which of the following is the first law for a closed system undergoing a cycle?
 (a) $\oint dW = \oint dQ$ (b) $\oint dW = -\oint dQ$ (c) $\oint dW = \oint dQ$ (d) None of the mentioned
- (iii) A heat engine produces an output of 12 kW by rejecting 23 kJ/sec of heat to a sink at a particular temperature. The efficiency of the engine will be (a) 33.4 % (b) 65.7% (c) 34.3% (d) 52.17%
- (iv) In a throttling device, what do we get as SFEE when changes in PE and KE are taken zero?
 (a) $dQ/dm \neq 0$ (b) $dW/dm \neq 0$ (c) $h_1 = h_2$ (d) none of the mentioned
- (v) Coefficient of performance(COP) is defined as
 (a) heat leakage/work input (b) work input/heat leakage
 (c) refrigeration effect/work input (d) work input/latent heat of condensation
- (vi) What is the dryness fraction (x) for saturated water, when water just starts boiling?
 (a) x = 0 (b) x = 1 (c) x = 0.9 (d) x = 0.5
- (vii) Entropy is a
 (a) path function, intensive property (b) path function, extensive property
 (c) point function, intensive property (d) point function, extensive property
- (viii) Triple point of a pure substance on P-V diagram is represented by a
 (a) point (b) line (c) curve (d) triangle
- (ix) At a pressure below the triple point line,
 (a) the substance cannot exist in the liquid phase
 (b) the substance when heated transforms from solid to vapour
 (c) both of the mentioned (d) none of the mentioned
- (x) The amount of heat transferred to convert unit mass of solid to vapour or vice versa is called as
 (a) latent heat of vapourization (b) latent heat of fusion
 (c) latent heat of sublimation (d) specific heat
- (xi) The function of a nozzle is to?
 (a) decrease the velocity of a fluid at the cost of its pressure gain
 (b) increase the velocity of a fluid at the cost of its pressure drop
 (c) increase the velocity of a fluid and also its pressure
 (d) none of the mentioned.
- (xii) A closed system receives 60 kJ of heat but its internal energy decreases by 30 kJ. Then the work done by the system is (a) - 90 kJ (b) -30 kJ (c) 30 kJ (d) 90 kJ
- (xiii) Mollier diagram is a plot of
 (a) temperature and entropy (b) enthalpy and entropy
 (c) pressure and enthalpy (d) pressure and volume
- (xiv) What is the function of economizer?

- (a) It is used to heat the feed water (b) It is used to improve the quality of steam
 (c) It is used to heat air being supplied for combustion
 (d) It separates suspended water particles from steam
- (xv) A mass of gas is compressed in a quasi-static process from 80 kPa, 0.1 m^3 to 0.4 MPa, 0.03 m^3 in a process following $PV^n = \text{const.}$ The index of compression is
 (a) 1.23 (b) 1.337 (c) 1.4 (d) 1.46

GROUP-B

Answer any Five (05) questions.

2. a) Define the following terms related to a thermodynamic system: Path of a process, Extensive property and & Point function. ①
 What do you understand by property of a system? $[(1 \times 3) + 1]$
- b) Prove that for a reversible adiabatic process, $PV^\gamma = \text{constant}$. ① (4)
3. a) A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C , enthalpy 2785 kJ/kg , velocity 33.3 m/s and elevation 3 m . The steam leaves the turbine at the following state: Pressure 20 kPa , enthalpy 2512 kJ/kg , velocity 100 m/s , and elevation 0 m . Heat is lost to the surroundings at the rate of 0.29 kJ/s . If the rate of steam flow through the turbine is 0.42 kg/s , what is the power output of the turbine in kW? (6) ②
- b) State Clausius inequality. (2)
4. a) State Charles' law. (1)
 b) What is PMM-I? Is it possible to devise a PMM-I? Justify your answer. $(1+1+1)$
- c) A house requires $2 \times 10^5 \text{ kJ/h}$ for heating in winter. Heat pump is used to absorb heat from cold air outside in winter and send heat to the house. Work required to operate the heat pump is $3 \times 10^4 \text{ kJ/h}$. Determine (i) Heat abstracted from outside and (ii) Co-efficient of performance. $(2+2)$
5. a) Derive the relationship between temperature and volume for an ideal gas, if it undergoes a polytropic process. <https://www.pyqonline.com> (2)
- b) A gas undergoes a thermodynamic cycle consisting of the following processes:
 • Process 1-2: Constant pressure $p = 1.4 \text{ bar}$, $V_1 = 0.028 \text{ m}^3$, $W_{12} = 10.5 \text{ kJ}$
 • Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$
 • Process 3-1: Constant volume, $U_1 - U_3 = -26.4 \text{ kJ}$. There are no significant changes in KE and PE.
 (i) Sketch the cycle on a p-V diagram (ii) Calculate the net work for the cycle in kJ. ①
 (iii) Calculate the heat transfer for process 1-2 (iv) Show that $\Sigma Q = \Sigma W$. $(1+2+1+2)$
6. a) Draw the phase equilibrium diagram on p-v & T-s coordinates considering water as a pure substance. Indicate all relevant constant property lines in the plot. (2+2)
- b) Find the specific enthalpy and specific entropy of steam when the pressure is 2 MPa and the specific volume is $0.09 \text{ m}^3/\text{kg}$. (2+2)
7. a) Derive the expression, $h = 353H \left[\frac{1}{T_a} - \left(\frac{m+1}{m} \right) \frac{1}{T_f} \right]$ of natural draught in m of water column. (all the symbols have their usual meanings). (5)
- b) Draw a neat sketch of a horizontal shell, water tube boiler and label its components. (3)
8. 2 kg of an ideal gas occupies a volume of 0.3 m^3 at 10 bar pressure and 500 K temperature. When this gas expands polytropically $pV^{1.2} = C$, the internal energy decreases by 300 kJ . Presuming adiabatic exponent $\gamma = 1.4$, determine: (i) characteristic gas constant, (ii) final temperature and pressure of gas and (iii) heat and work interactions across the system boundary. $[1+(2+2)+(2+1)]$
9. a) 10 kg of wet steam at a pressure of 0.2 MPa is contained in a rigid tank of volume 6.058 m^3 . The tank is heated until the steam becomes dry saturated. Determine the final pressure and the heat transfer to the tank. (3+3)
- b) What do you understand by degree of superheat? What is the value of pressure and temperature at

critical point in case of water?

(1+1)

10. a) Why accessories are essential in boiler? Name two accessories used in boiler and state the function of steam stop valve and fusible plug. ① (1+1+1+1)
- b) A boiler evaporates 5 kg of water per kg of coal burnt on the fire grate in the combustion chamber. The temperature of feed water is 30°C and the steam is raised at 10 bar absolute pressure. Calculate the equivalent evaporation "from and at 100°C " per kg of coal and the efficiency of boiler, if the steam produced is 0.95 dry. Take calorific value of coal as 29800 kJ/kg. (2+2)
-

ADVANCED STRENGTH OF MATERIALS

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and to be answered first.**This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.****Answer any five questions from the rest.**

1. Answer the following questions (any ten):

1×10=10

Fill in the blanks:

- i) Bending stress at N.A. is _____.
- ii) The section modulus of a beam having width b and depth d is given by _____.
- iii) The ratio of the polar moment of inertia to the radius of the shaft is called _____.
- iv) Force required to deflect a spring through unit deflection is called _____.
- v) In a strained body, maximum number of principal planes may be _____.
- vi) The maximum amount of strain energy that can be stored in a body without deformation is called _____.
- vii) A shaft is said to be in pure in pure tension, if it is subjected to _____ only.

State whether the following statements are True or False:

- viii) The stiffness of spring is also called spring constant.
- ix) The product of $G \times I_p$ is called torsional rigidity.
- x) The neutral axis of a beam is subjected to maximum tensile stress.
- xi) Core or kernel is the area within which the resultant of the load may be applied to avoid tensile stress.
- xii) Polar moment of inertia of a circular section of diameter D is $\pi/32 D^4$
- xiii) If section modulus of a beam is increase, the bending stress will increase.
- xiv) For a beam with rectangular section, maximum shear stress occurs at neutral axis.
- xv) If a coil spring is cut into two halves, deflection of the spring will be half.

2. What is simple bending? What are the assumptions made in the simple bending? 2+3
3. A steel plate 12 mm thick is bent around a drum. If the maximum stress induced in the plate due to bending be 400 N/mm^2 , find the radius of the drum. Take $E = 2 \times 10^5 \text{ N/mm}^2$. 5
4. Strain energy of two elastic bars subjected to gradual loading having cross-sectionals area are 400 mm^2 and 800 mm^2 are same. Establish a relation between stresses of two bars. 5
5. An I-section beam of flanges $20 \text{ cm} \times 2 \text{ cm}$ and web $30 \text{ cm} \times 1 \text{ cm}$ is acted on by a shear force of 150 kN determine: i).the maximum and minimum shearing stress in the web. ii).the maximum shear stress in the flange. 5
6. The principal stress at a point in the section of a boiler shell are 80 N/mm^2 and 40 N/mm^2 . Find the normal, tangential and resultant stress across a plane through the point inclined at 50° to the plane carrying 80 N/mm^2 stress. 5

7. Deduce torsion equation i.e. $\frac{T}{I_p} = \frac{\sigma_s}{R} = \frac{G\theta}{L}$ for solid shaft. 5
8. A solid shaft of 80 mm diameter is to be replaced by a hollow shaft of external diameter of 100 mm. 5 Determine the internal diameter of hollow shaft if the same power is to be transmitted by the both shafts at the same angular velocity and shear stress. 5
9. A closed coiled helical spring having the mean coil diameter is 12 times than the wire diameter. The spring is to absorb 30000 N cm of energy with an extension of 15 cm. The maximum shear stress set up in the spring is not to exceed 10000 N/cm². Determine – (a) mean coil diameter, (b) number of turns. 5
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March 2021

ADVANCED STRENGTH OF MATERIALS

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any ten): 1x10

State whether the following statements are True or False:

- i) The strength of the beam mainly depends on the bending moment.
- ii) Strain energy is set up in elastic material.
- iii) Moment of resistance of beam is independent of its section modulus.
- iv) If the section modulus of a beam is increased, the bending stress of the beam will be decrease.
- v) Proof resilience is the maximum strain energy upto elastic limit.
- vi) Spring index is the ratio of mean coil diameter to the diameter of spring.
- vii) For a circular section shear stress distribution is parabola.
- viii) The unit of section modulus and area are same.

Fill in the blanks:

- ix) The bending equation is _____.
- x) The bending stress on a beam section is maximum at _____.
- xi) For a T-section beam, maximum shear stress occurs at _____.
- xii) The angle between the principle planes is _____.
- xiii) The radius of Mohr's stress circle indicates maximum _____ stress.
- xiv) Strain energy is set up in _____ material.
- xv) Proof resilience per unit volume is called _____.
- xvi) The unit of modulus of elasticity is same as of _____.

2. An I-section beam of flanges 20 cm × 2 cm and web 30 cm × 1 cm is acted on by a shear force of 150 kN determine: i).the maximum and minimum shearing stress in the web. ii).the maximum shear stress in the flange. 5
3. The principal stress at a point in the section of a boiler shell are 80 N/mm² and 40 N/mm². Find the normal, tangential and resultant stress across a plane through the point inclined at 50° to the plane carrying 80 N/mm² stress. 5
4. Deduce torsion equation i.e. $\frac{T}{I_p} = \frac{\sigma_s}{R} = \frac{G\theta}{L}$ for solid shaft. 5
5. A solid shaft of 80 mm diameter is to be replaced by a hollow shaft of external diameter of 100 mm. Determine the internal diameter of hollow shaft if the same power is to be transmitted by the both shafts at the same angular velocity and shear stress. 5
6. A closed coiled helical spring having the mean coil diameter is 12 times than the wire diameter. The spring is to absorb 30000 N cm of energy with an extension of 15 cm. The maximum shear stress set up in the spring is not to exceed 10000 N/cm². Determine – (a) mean coil diameter, (b) number of turns. 5
7. What is simple bending? What are the assumptions made in the simple bending? 2+3

8. A steel plate 12 mm thick is bent around a drum. If the maximum stress induced in the plate due to bending be 400 N/mm^2 , find the radius of the drum. Take $E = 2 \times 10^5 \text{ N/mm}^2$. 5
9. Strain energy of two elastic bars subjected to gradual loading having cross-sectional area are 400 mm^2 and 800 mm^2 are same. Establish a relation between stresses of two bars. 5
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STRENGTH OF MATERIALS*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten): 1 x 10
- i. Spring is used to- a) Absorb shock, b) Accelerate speed of a vehicle, c) Both (a) & (b), d) None of the above.
 - ii. The property of a material by virtue of which it can be rolled into plate known as-
a) Ductility b) toughness c) malleability d) none of these.
 - iii. Neutral axis passes through-
a) The C.G. of the section of the beam b) the top layer of the beam
c) The bottom layer of the beam d) none of the above
 - iv. For circular section of diameter 'd', the section modulus is given by _____.
 - v. If diameter of a long column is reduced by 20%, the percentage reduction of Euler buckling load is -
(a) 59, (b) 49, (c) 8, (d) 36.
 - vi. Slope and deflection of a beam can be determined by-
a) Double integration method b) Moment area method
c) Macaulay's method d) any of the above
 - vii. A strut is a member which carries - (a) tensile load, (b) shear load, (c) compressive load, (d) both tensile and compressive load.
 - viii. Poisson's ratio for any material cannot be more than-
a) 0.3, b) 0.5, c) 0.8, d) 1
 - ix. Deformation of a body under tensile load P of cross sectional area A of Length L having Young's modulus E is given by-
a) $\frac{PE}{AL}$ b) $\frac{AE}{PL}$
c) $\frac{PL}{AE}$ d) none of these
 - x. Hooke's law holds good upto the - (a) elastic limit, (b) plastic limit, (c) yield point, (d) proportional limit.
 - xi. Unit of modulus of rigidity - a) N/m^2 b) N/mm^2 c) both a and b d) none of these
 - xii. Bending moment diagram of uniformly distributed force will be-
a) Linear b) Inclined
c) Non-linear d) Horizontal
 - xiii. Percentage of elongation is the measure of-
a) Ductility, b) Malleability, c) Plasticity, d) Elasticity
 - xiv. If diameter of a shaft is doubled the power transmitted capacity will be - (a) same, (b) either twice or half, (c) four times, (d) eight times.

- xv. Shafts are designed on the basis of-
(a) strength (b) rigidity (c) either of the above (d) either (a) or (b)

2. Fill in the blanks (any ten):

1x10

- i. The maximum bending moment (M) caused by the total uniformity distributed load (W) over a simply supported span (L) is given by _____.
- ii. The ratio of crippling load to working load is called _____.
- iii. The angle of twist is _____ proportional to the twisting moment
- iv. Reaction at free end of a cantilever beam is _____.
- v. A beam can be made of uniform strength by keeping the _____ uniform, and varying the _____.
- vi. Unit of radius of gyration is _____.
- vii. The ratio of lateral strain to linear strain is known as _____.
- viii. When a SF at a point is zero, then BM at that point is _____.
- ix. Permanent set occurs when stress is beyond _____ limit.
- x. For a beam with rectangular section, the ratio of maximum shear stress to average shear stress is _____.
- xi. Unit of coefficient of thermal expansion is _____.
- xii. _____ is a graphical method to determine various stresses on an oblique plane of a strained material.
- xiii. The combined effect of external forces acting on a body is called a _____.
- xiv. The strain at right angles to the direction of applied load is known as _____.
- xv. Percentage elongation is a measure of _____.

3. Answer the following questions (any ten):

1x10

- i. Define brittle material.
- ii. What is the helix angle of helical spring?
- iii. Show uniformly varying load on a beam.
- iv. Write the mathematical expression of Temperature stress.
- v. Write down the relation between bulk modulus & compressibility.
- vi. What do you mean by polar modulus?
- vii. What is the expression of Torque transmitted by a solid shaft of diameter d?
- viii. Write the mathematical expression of Polar Modulus of a shaft.
- ix. What is torsional rigidity?
- x. What is the maximum value of S.F, in case of simply supported beam of span l loaded with a point load W at its mid span?
- xi. Describe bending stress.
- xii. What is meant by section modulus?
- xiii. What is law of transmissibility?
- xiv. What is the factor of safety?
- xv. What is SI unit of UDL?

Group B

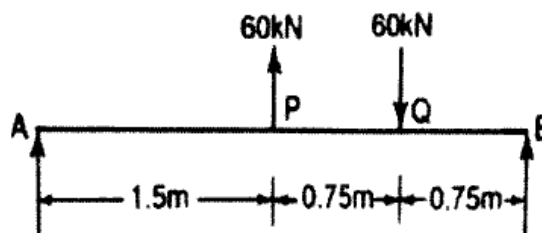
4. Answer the following questions (any six):

2x6

- i. What is value of bulk modulus of a material having $E=250\text{GPa}$ and $G=80\text{GPa}$?
- ii. Draw a stress strain diagram for ductile materials and indicate all salient features on it.
- iii. Write the equation of torsion and give the names of each terms.
- iv. Define stiffness of a spring.
- v. Write down the difference between deflection & bending moment.
- vi. What is Pure Bending?
- vii. Define SFD and BMD.
- viii. State the perpendicular axis theorem.
- ix. What do you mean by slenderness ratio?
- x. What are the application areas of spring?

Group C

5. Answer the following question (any one): 1x6
- i. What is moment of inertia. What is the difference between mass moment of inertia & area moment of inertia? 2+4
 - ii. Discuss the stress strain behavior of ductile material. Establish the relation $E = 2G(1 + \mu)$ where E is Young's modulus, G is modulus of rigidity and μ is Poisson's ratio. 2+4
 - iii. Find the MI of a Triangular area with respect to the axis passing through the base of the triangle, where, base=b and height =h. 6
6. Answer the following question (any one): 1x6
- i. What are the assumptions made in the Euler's formula? Explain the term effective length of a column. 4+2
 - ii. A cantilever 2m long carries point loads of 4kN, 5kN, 3kN at distances 1m, 1.5m, and 2m respectively from the fixed end sides. Draw the shear force and bending moment diagram for the beam. 6
 - iii. A T beam 200mm*150mm*10mm is used over a span of 3 meter It carries a uniformly distributed load of 12000 N per meter run. Find the position of the neutral axis and the maximum stress induced in the beam due to bending. Also state the nature of maximum bending stress. 6
7. Answer the following question (any one): 1x6
- i. A solid circular shaft of 100 mm diameter is transmitting 120 kW at 150 r.p.m. Find the intensity of shear stress in the shaft. <https://www.wbscteonline.com> 6
 - ii. What do you mean by equivalent stiffness of helical spring? Draw sketches showing series connection and parallel connection of helical springs. 2+4
 - iii. A beam of 3m span is loaded as shown in the fig 6. Calculate the deflection at P and Q and slope at A. Take the value of $EI = 840 \text{ KN-m}^2$ 6



March 2023

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): 1x20
- i) The material which shows little or no plastic deformation is called – (a) ductile material (b) brittle material (c) malleable material (d) plastic material.
 - ii) The point at which there is increase in strain with no appreciable increase in stress is known as – (a) Breaking point (b) Yield point (c) Elastic limit (d) Proportional limit.
 - iii) Ratio of Hoop stress and longitudinal stress of a material is equal to – (a) 1 (b) $\frac{1}{2}$ (c) 2 (d) none of these.
 - iv) Temperature stress developed in a bar depends upon – (a) co-efficient of linear expansion (b) change of temperature (c) Young's modulus (d) all of these.
 - v) The area moment of inertia is often called _____ moment of area. – (a) first (b) second (c) third (d) fourth
 - vi) Moment of Inertia of a triangular lamina of height 'h' about its base 'b' is – (a) $\frac{b \times h^3}{12}$ (b) $\frac{b \times h^3}{36}$ (c) $\frac{h \times b^3}{12}$ (d) none of these.
 - vii) If I_{OX} and I_{OY} be moments of inertia of a lamina about mutually perpendicular axes OX and OY in the plane of the lamina and I_{OZ} be the moment of inertia of the lamina about an axis OZ normal to the lamina and passing through the point of intersection of the axis OX and OY then $I_{OZ} = I_{OX} + I_{OY}$. The above theorem is known as – (a) theorem of parallel axes (b) theorem of perpendicular axes (c) three moments theorem (d) none of these.
 - viii) A simple supported beam of length 10 m carries uniformly distributed load of 10 kN/m over the entire span. The magnitude of each support reaction will be – (a) 25 kN (b) 30 kN (c) 50 kN (d) 100 kN.
 - ix) The maximum bending moment of a cantilever beam appears at – (a) fixed end (b) free end (c) mid-span (d) none of these.
 - x) If a beam is loaded with UDL, the nature of bending moment diagram will be – (a) straight horizontal line (b) straight inclined line (c) parabolic (d) cubic.
 - xi) At the point of contra flexure, the value of bending moment will be – (a) maximum (b) minimum (c) zero (d) none of these.
 - xii) The bending stress at neutral axis is – (a) maximum (b) minimum (c) zero (d) tensile in nature.
 - xiii) The section modulus of a circular section of diameter d is – (a) $\frac{\pi d^2}{32}$ (b) $\frac{\pi d^3}{32}$ (c) $\frac{\pi d^3}{64}$ (d) $\frac{\pi d^4}{64}$.
 - xiv) Neutral axis passes through – (a) the top layer of the beam (b) the bottom layer of the beam (c) the CG of the beam (d) none of these.

- xv) Maximum deflection of a beam occurs where the slope of the beam is – (a) maximum (b) 45° (c) 90° (d) 0° .
- xvi) The maximum deflection of a cantilever beam appears at – (a) fixed end (b) free end (c) mid-span (d) none of these.
- xvii) Doubling the span of beam will result in increase in deflection of the beam by – (a) two times (b) four times (c) eight times (d) none of these.
- xviii) Euler's formula for crippling load is best suitable for slenderness ratio more than – (a) 50 (b) 70 (c) 80 (d) 100
- xix) The crippling load of a column depends on – (a) cross-sectional area (b) modulus of rigidity (c) slenderness ratio (d) all of these.
- xx) The ratio of actual length to the equivalent length of a column having one end fixed and other end free is – (a) 1:1 (b) $\sqrt{2}$:1 (c) 1:2 (d) $1:\sqrt{2}$.
- xxi) The strength of a hollow shaft for the same length, material and weight is _____ a solid shaft. – (a) less than (b) more than (c) equal to (d) none of these.
- xxii) In pure torsion, the shaft gets deformed due to – (a) bending moment only (b) axial load only (c) twisting moment only (d) all of these.
- xxiii) For a hollow circular shaft torsional shear stress at the centre is – (a) maximum (b) zero (c) minimum (d) none of these.
- xxiv) The predominant effect of an axial tensile force on a helical spring is – (a) bending (b) compression (c) twisting (d) tension.
- xxv) Two helical springs of stiffness K is attached in parallel. The stiffness of the equivalent system will be – (a) 2K (b) K/2 (c) 2K (d) 4K

Group B

2.
 - i) Draw the stress-strain diagram of Mild Steel and locate all the salient points in the diagram.
 - ii) A steel wire 2 m long and 3 mm in diameter is extended by 0.75 mm when a weight W is suspended from the wire. If the same weight is suspended from a brass wire, 2.5 m long and 2 mm in diameter, it is elongated by 4.64 mm. Determine the modulus of elasticity of brass if that of steel be $2.0 \times 10^5 \text{ N/mm}^2$. 3+5
3.
 - i) Define volumetric strain and bulk modulus of a material.
 - ii) A copper bar of diameter 25 mm and length 2 m is rigidly fixed at both ends. Find the thermal stress induced in the bar if the temperature is reduced by 40°C . Take $E = 100 \text{ GPa}$, $\alpha = 16 \times 10^{-6}/^\circ\text{C}$. 2+6
4.
 - i) State the "Theorem of parallel axis".
 - ii) Show that the moment of inertia of a circular section with respect to an axis passing through its centre of gravity is $\pi D^4/64$, where D is the diameter of the circular section. 2+6
5.
 - i) Write down the relationship between shear force and bending moment and explain its significance.
 - ii) A simply supported beam 2.5 m long is carrying two point loads of 2 kN and 4 kN at a distance of 1 m & 1.5 m from left end respectively. Draw the SF & BM diagram. 2+6
6.
 - i) What are the assumptions in Simple Bending Theory?
 - ii) A simply supported beam of span 5 m has a cross section 150 mm x 250 mm. If the permissible stress is 10 N/mm^2 , find the maximum intensity of uniformly distributed load it can carry. 2+6
7.
 - i) Write the formula for maximum slope of a cantilever beam with a point load at free end.
 - ii) A cantilever beam of 2 m length having width 100 mm and depth 120 mm is subjected to a point load of 30 kN at its free end. Find the maximum slope and deflection at free end if the Young's Modulus of the beam material is 200 GPa. 1+7

8. i) What is slenderness ratio?
ii) Find the Euler's crippling load for a hollow cylindrical steel column of 38 mm external diameter and 2.5 mm thick. Take length of the column as 2.3 m and hinged at its both ends. Take $E = 205 \text{ GPa}$. 2+6
9. i) Write down the equation of Pure torsion and explain each term.
ii) A solid circular shaft transmits 100 kW power at 200 r.p.m. Calculate the shaft diameter if twist in the shaft is not to exceed 1° in 2 metres of the shaft length and the shearing stress is limited to 70 N/mm^2 . Take modulus of rigidity $= 8 \times 10^4 \text{ N/mm}^2$. 2+6
10. i) What is Spring index?
ii) A close coil helical spring to carry a load of 250 N, if the mean coil diameter is to be 8 times that of the wire, calculate these diameters. Take maximum shear stress as 70 MPa . 2+6
-

**STRENGTH OF MATERIALS**

Time Allowed: 3 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script.
In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten):
2×10=20

- i) The strength of a beam depends on –
(a) bending moment (b) its weight ✓
(c) section modulus (d) shape of the section
- ii) If a coiled spring is cut into two halves, stiffness of the spring will be –
(a) double, (b) half, (c) same, (d) quadruple.
- iii) A beam is a structural element subjected to _____ load.
(a) axial, (b) transverse, (c) seismic, (d) none of these.
- iv) If a rod expands freely due to heating, it will develop –
(a) thermal stress, (b) bending stress, (c) compressive stress, (d) no stress.



- v) If diameter of a shaft is doubled the power transmitted capacity will be –
(a) same, (b) either twice or half, (c) four times, (d) eight times. ✓
- vi) Polar moment of inertia of a circular area of diameter 'D' –
(a) $\frac{\pi}{64} D^4$, (b) $\frac{\pi}{32} D^4$, (c) $\frac{\pi}{8} D^4$, (d) —
- vii) A strut is a member which carries –
(a) tensile load, (b) shear load, (c) compressive load, (d) both tensile and compressive load. ✓
- viii) The maximum deflection of a simply supported beam of length 'l' with a uniformly distributed load of 'w' per unit length is –
(a) —, (b) —, (c) —, (d) —
- ix) The shape of shear force diagram for a cantilever with a point load at the free end will be –
(a) rectangular, (b) curve, (c) square, (d) triangular.



- x) The intensity of shear stress at a section is _____ to the distance of the section from the axis of the shaft.
(a) inversely proportional, (b) directly proportional, (c) equal, (d) parallel
- xi) The relationship between Young's modulus (E), Bulk modulus (K) and Poisson's ratio (ν) is given by –
(a) $E = 3K(1 - 2\nu)$ (b) $E = 2K(1 - 2\nu)$
(c) $E = 3K(1 + 2\nu)$ (d) $E = 2K(1 - 3\nu)$
- xii) If diameter of a long column is reduced by 30%, the percentage reduction of Euler buckling load is – (a) 59, (b) 49, (c) 8, (d) 76.
- xiii) The slenderness ratio is the ratio of effective length to least _____.
(a) ultimate load (b) actual length
(c) radius of gyration (d) factor of safety
- xiv) When the helical extension spring is subjected to axial tensile force, the type of stress induced in the spring wire is –
(a) compressive stress, (b) torsional shear stress, (c) tensile stress, (d) bending stress.
- xv) In a thin cylindrical shell subjected to an internal pressure (p), the ratio of hoop stress to longitudinal stress is – (a) $1/2$, (b) 1, (c) 2, (d) 4



GROUP-B

Answer any Five (05) questions.

2. a) Illustrate stress-strain curve for ductile (M.S) material & discuss salient points.
b) The diameter of a water main is 2 m and the thickness of its wall is 20 mm. Determine the circumferential and longitudinal stresses set up in the pipe material when water is introduced in it at a pressure of 1.5 N/mm^2 . Take, Young's modulus (E) = 200 kN/mm^2 .
4+4
3. a) Deduce the formula for the elongation of a uniform cross-sectional bar under the action of its own weight.
b) Find the force required to punch a hole of 20 mm in diameter in a metal plate of 10 mm thickness. If the ultimate strength of the plate metal in shear is 300 N/mm^2 , determine the compressive stress set up in the punch.
5+3



3223770

4.

Find the moment of inertia of the I-section shown in Fig.1 about the horizontal axis passing through the centroid of the section.

8

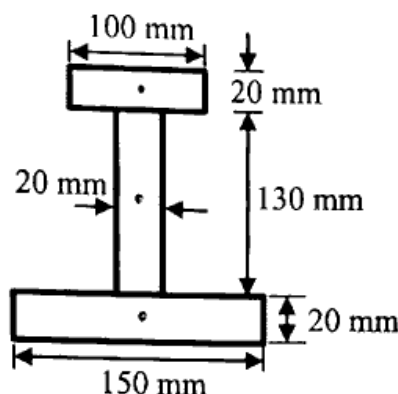


Fig.1

5.

Draw shear force and bending moment diagrams for the beam with loads as shown in the Fig.2. Show all important points in the diagrams.

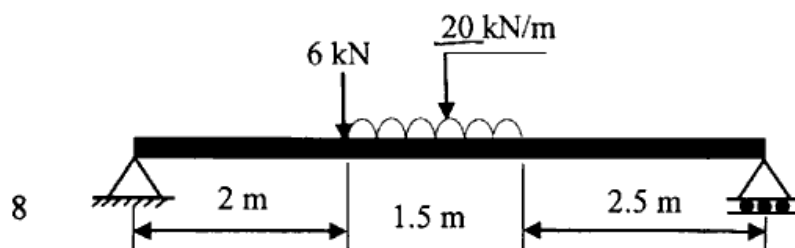


Fig. 2



6. a) What do you mean by pure bending?
 b) Deduce the bending equation: $\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$ with usual notations. 2+6
7. a) A cantilever 2 m long carries a uniformly distributed load of 30 kN per metre run over its entire length. Taking, Young's modulus (E) = 2×10^4 kN/cm² and moment of inertia of the section of the cantilever about the neutral axis = 10,000 cm⁴. Determine, (i) deflection at the free end, (ii) slope at the free end.
 b) What is flexural rigidity? 6+2
8. a) What is the difference between a column and a strut?
 b) A hollow column of 200 mm external diameter and 160 mm internal diameter is used as a column of 4.5 m length. Calculate the Rankine crippling load when the column is pinned at both ends.
 Take allowable stress as 300 MPa and Rankine's constant as 1/7500. 3+5



3223770

9. a) The diameter of a solid shaft is 250 mm. Find out the maximum safe torque which the shaft can transmit if the allowable shear stress is 4600 N/cm^2 and the maximum angle of twist is 0.25° per metre length. Take, Modulus of rigidity (G) = $8 \times 10^6 \text{ N/cm}^2$. If the shaft rotates at 200 r.p.m, what maximum power can be transmitted by the shaft? 6+2
- b) What do you mean by polar modulus?
10. a) Deduce a formula for the equivalent stiffness of helical springs arranged in series connection and parallel connection.
- b) A closed coiled helical spring having the mean coil diameter is 60mm and the diameter of the wire with which the spring is made is 5 mm. The axial load carried by the spring is 200 N. The spring is made of 20 effective turns. Taking, Modulus of rigidity (G) = 80 GPa, determine – 4+4
- (i) maximum shear stress developed in the spring, (ii) deflection of the spring.



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JANUARY 2024

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

i. Which point on the stress strain curve occurs after the ultimate point?

- a) Last point b) Breaking point
c) Elastic limit d) Material limit

ii. The hinged end is also known as _____

- a) Fixed end b) Pinned end
c) Rigid end d) Free end

iii. The slenderness ratio is the ratio of effective length to least _____

- a) Ultimate load b) Actual length
c) Radius of gyration d) Factor of safety

iv. _____ is half the circumferential stress.

- a) Hoop stress b) Longitudinal stress
c) Fluid stress d) Transverse stress

v. What are the units of Polar modulus?

- a) mm^3 b) mm^2
c) mm d) mm^4

vi. U.D.L stands for?

- a) Uniformly diluted length b) Uniformly developed loads
c) Uniaxial distributed load d) Uniformly distributed loads

vii. Units of U.D.L?

- a) KN/m b) KN-m
c) $\text{KN-m} \times \text{m}$ d) KN

viii. At the Point of contraflexure, what is the value of bending moment?

- a) one b) zero
c) three d) infinity

ix. When SF is zero, the bending moment is _____

- a) Zero b) Maximum
c) Very difficult to say d) Minimum

x. What is the moment of inertia of a rectangular section about an horizontal axis through C.G?

- a) $bd^3/6$ b) $bd^2/12$
c) $b^2d^2/12$ d) $bd^3/12$

xi. What is the section modulus (Z) for a rectangular section?

- a) $bd^2/6$ b) $a^3/6$
c) BD^3-bd^3 d) D^4-d^4

xii. On bending of a beam, which is the layer which is neither elongated nor shortened?

- a) Axis of load b) Neutral axis
c) Center of gravity d) None of the mentioned

xiii. In cantilever beam the deflection occurs at _____

- a) Free end b) Point of loading
c) Through out d) Fixed end

xiv. The angle of twist can be written as _____

- a) TL/J b) CJ/TL
c) TL/CJ d) T/J

xv. Shear force is unbalanced _____ to the left or right of the section.

- a) Horizontal force b) Vertical force c) Inclined force d) Conditional force

Group-B

2. Draw the stress-strain diagram for a mild steel specimen. Show the salient points in the diagram (8)

3. A cantilever beam AB, 2m long carries a uniformly distributed load of 1.5 kN/m over a length of 1.6 m from the free end. Draw shear force and bending moment diagrams for the beam. (8)

4. Find the moment of inertia of a T-section with flange as 150 mm x 50 mm and web as 150 mm x 50 mm about X-X and Y-Y axes through the centre of gravity of the section. (8)

5. Calculate the maximum torque that a shaft of 225 mm diameter can transmit, if the maximum angle of twist is 1° in a length of 1.5 m. Take $C = 70 \times 10^3 \text{ N/mm}^2$ (8)

6. A steel rod 5 m long and of 40 mm diameter is used as a column, with both end hinged. Determine the crippling load by Euler's formula. Take $E = 200 \times 10^3 \text{ N/mm}^2$ (8)

7. A closely-coiled helical spring of round steel wire 5 mm in diameter having 12 complete coils of 50 mm mean diameter is subjected to an axial load of 100 N. Find the deflection of the spring and the maximum shearing stress in the material. Modulus of rigidity (C) = $80 \times 10^3 \text{ N/mm}^2$ (8)

8. A steam boiler of 800mm diameter is made up of 10 mm thick plates. If the boiler is subjected to an internal pressure of 2.5 MPa. Find the circumferential and longitudinal stresses induced in the boiler plates. (8)

9. Write short notes on the followings (Any four) a) Elasticity b) Malleability c) Fatigue d) Creep e) Poisson's Ratio f) Brittleness (4×2 = 8)

10. Prove that $T/J = C.\theta/L$ for a solid circular shaft. What are the assumptions made in derivation of torsion equation. (5 + 3 = 8)

STRENGTH OF MATERIALS

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for the right answer and 1 mark is allotted for the correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20
 - (i) The stress at which elongation of a material is quite large as compared to the increase in load is known as-
(a) Ultimate point (b) Yield point (c) Elastic limit (d) Rupture point.
 - (ii) For a Poisson's ratio 0.4 for a material, the ratio of the shear modulus of elasticity to modulus of elasticity is- (a) 14/5 (b) 5/7 (c) 7/5 (d) 5/14.
 - (iii) Which of the following increases bending stress in a beam?
(a) Increase in moment of inertia (b) Increase in section modulus (c) Decrease in distance from neutral axis (d) Decrease in moment of inertia.
 - (iv) If a simple supported beam is loaded with a concentrated load W at the centre of the span then its B.M diagram takes the shape of -
(a) Rectangle (b) Parabola (c) Triangle (d) Square.
 - (v) When a beam is subjected to a negative bending moment-
(a) Top fiber is in tension and bottom fiber is in compression (b) Top fiber is in compression and bottom fiber is in tension. (c) Minimum stress in top and bottom fiber (d) Both (a) and (b).
 - (vi) The M.O.I. of a circle of diameter D with respect to the axis which is perpendicular to its diameter and passing through the centre is -
(a) $\frac{\pi}{64} D^4$ (b) $\frac{\pi}{32} D^4$ (c) $\frac{\pi}{32} D^3$ (d) $\frac{\pi}{16} D^3$
 - (vii) In a simple supported beam carrying UDL ' w ' per unit length, the point of contraflexure -
(a) Lies in the centre of the beam (b) Lies at the ends of the beam (c) Depends upon the length of the beam (d) Does not exist.
 - (viii) If a shaft diameter is doubled, its torsional strength increases by-
(a) 2 times (b) 4 times (c) 8 times (d) 16 times.
 - (ix) The buckling load of a column for a given material depends upon-
(a) Poisson's ratio and slenderness ratio (b) Slenderness ratio and modulus of elasticity (c) Poisson's ratio and modulus of elasticity (d) Slenderness ratio and cross sectional area.
 - (x) The ratio of torsional strengths of solid to hollow shafts, both having outside diameter D and hollow having inside diameter $D/2$, is-
(a) 15/16 (b) 1/8 (c) 1/16 (d) 16/15.

- (xi) In the context of beam bending, flexural rigidity is -
 (a) A measure of the beam's resistance to shear stress (b) A measure of the beam's resistance to bending
 (c) A measure of the beam's ability to support compressive loads (d) A measure of the beam's ability to support tensile loads.
- (xii) A column is fixed in one end and other end is free and has a radius of gyration $K = 2.5\text{ cm}$. The actual length of the column is 1.2 m . The slenderness ratio of the column is -
 (a) 96 (b) 80 (c) 90 (d) 120.
- (xiii) If the flexural rigidity of a beam becomes double then the deflection of the beam-
 (a) Increases 2 times (b) Becomes half (c) decreases 4 times (d) remains same.
- (xiv) Four springs each having stiffness k are arranged in series, then the equivalent stiffness will be-
 (a) $4k$ (b) $k/2$ (c) $2k$ (d) $k/4$.
- (xv) Which of the following end condition gives the lowest Euler's buckling strength for a column?
 (a) Both ends fixed (b) Both ends pinned (c) One end fixed, other free (d) One end fixed, other hinged.

GROUP-B

2. (a) Draw the stress-strain curve for M.S material and discuss salient points in the diagram. (3)
 (b) A circular steel bar having three segments is subjected to various forces at different cross sections as shown in fig. 1. The diameter of the first, middle and last section of the bar is 30 mm , 25 mm and 30 mm respectively. Determine the total elongation of the bar. Take $E = 200\text{ GPa}$. [3+5]

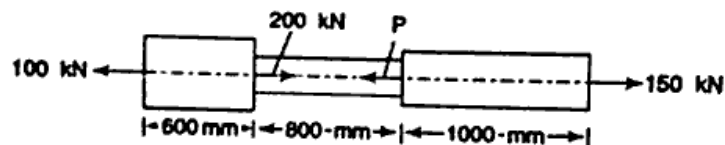


Fig. 1.

3. (a) A bar of 12 mm diameter and 2 m length is subjected to an axial load of 20 kN . The change in diameter is measured as 0.003 mm . If the modulus of rigidity of the bar is 80 GPa , determine (i) Poisson's ratio (ii) modulus of elasticity and bulk modulus.
 (b) No stress is developed in the rails when they are laid at 24°C . If the rails are 32 m long, calculate the stress developed in the rails at 80°C when it is allowed to expand 8 mm . Take co-efficient of linear expansion, $\alpha = 11 \times 10^{-6}/^\circ\text{C}$ and $E = 205\text{ GPa}$. [5+3]
4. Determine the moment of inertia of the composite shape given in the fig. 2 about x-axis and y-axis. [8]

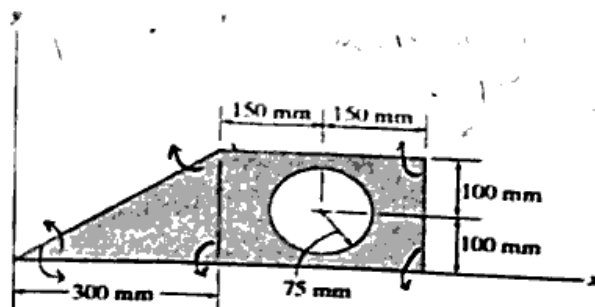


Fig. 2.

5. Draw shear force and bending moment diagrams for the beam as shown in the fig.3. Determine the maximum bending moment and its location. [8]

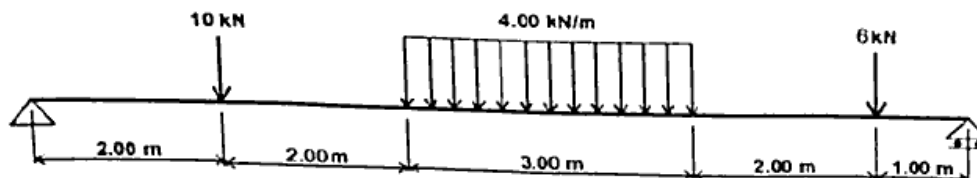


Fig.3.

- 6.(a) What do you mean by section modulus of a beam? (2)

(b) A cast iron beam of T section as shown in fig.4 is simply supported on a span of 8m. The beam carries a uniformly distributed load of 1.5kN/m length on the entire span. Determine the maximum tensile and maximum compressive stress. [2+6]

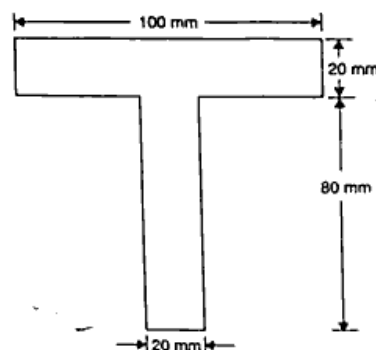


Fig.4.

7. (a) State the four assumptions in theory of bending.

(b) A beam of uniform rectangular section 20cm wide and 30cm deep is simply supported at its ends. It carries a uniformly distributed load of 9 kN/m run over the entire span of 5 m. If the value of E for the beam material is $1 \times 10^4 \text{ N/mm}^2$, find the slope at the supports and maximum deflection. [2+6]

8. (a) What is torsional stiffness?

(b) A hollow shaft with inner to outer diameter ratio $3/5$ is required to transmit 450kW at 120rpm with a uniform twisting moment. The shear stress in the shaft must not exceed 60 MPa and twist in a length of 2.5m must not exceed 1° . Calculate the least external and internal diameter of the shaft satisfying these conditions. Take $G = 80 \text{ GPa}$. [2+6]

9. (a) Draw the effective lengths of a column for various end conditions. (2)

(b) A hollow cylindrical column of length 6m having external diameter 200mm and thickness 25mm is fixed at both ends. Determine the ratio of Euler's critical load to Rankine's critical load. Take $E = 80 \text{ GPa}$; allowable stress as 550 MN/m^2 and Rankine's constant as $1/1600$. [3+5]

10. (a) What are the different types of springs? State the functions of the springs.

(b) A closely coiled helical spring is to carry a load of 500 N. The spring index is 10. If the maximum shear stress in the material of the spring is to be 80 MPa, determine mean coil diameter and wire diameter of the spring. If the stiffness of the spring is 20 N per mm deflection, calculate the energy stored in the spring and the number of turns in the spring. Take $G = 80 \text{ GPa}$ [2+6]

MECHANICAL ENGINEERING DRAWING*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the loose sheet and questions of Group-B must be written in drawing sheet. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

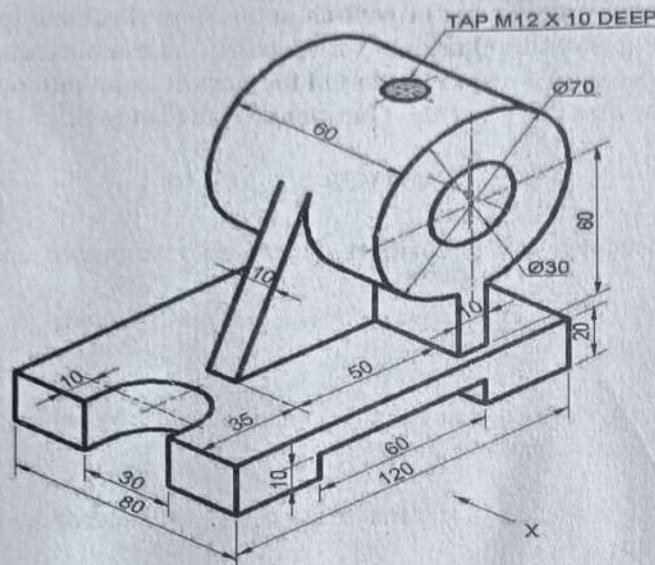
1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10
 - I. Detailed drawing of each part of a machine is called - (a) Assembly drawing, (b) part drawing, (c) patent drawing, (d) tabular drawing
 - II. Tolerance is the difference between - (a) maximum and minimum size (b) maximum and basic size (c) basic size and minimum size (d) none of these
 - III. Sectional views reveal - a) Inner details, b) external features, c) overall size of the object d) none of these.
 - IV. Basic Hole means - (a) Upper deviation is zero, (b) lower deviation is zero, (c) both a & b, (d) none of these
 - V. H7g6 - what type of fit is this? a) Clearance b) interference c) transition d) none of these.
 - VI. In a full sectional view the object is imagined to be cut off (a) one-third (b) one half (c) one fourth.
 - VII. A knuckle joint is a mechanical joint used to connect two rods which are under (a) tensile load (b) Compressive load (c) shear load (d) none of these
 - VIII. When a horizontal cylinder penetrates a vertical cone the intersecting surface will have appeared as (a) a straight line (b) curve (c) None of these
 - IX. A coupling which permits within certain limits relative rotation and variation in the alignment of shafts is called (a) Flexible coupling (b) Rigid coupling (c) Universal coupling (d) None of these
 - X. Radial-line development method are used to develop (a) Cylinder (b) Cone (c) Pyramid (d) both b & c
 - XI. Which of the following are part of the knuckle joint (a) pin (b) fork (c) eye (d) all of this
 - XII. Minimum no. of views necessary to determine the shape of the object are: (a) 1, (b) 2, (c) 3, (d) 4
 - XIII. Most common material for pulley is (a) cast iron (b) wood (c) rubber (d) brass
 - XIV. The angle between isometric axes is - (a) 30° , (b) 45° , (c) 90° , (d) 120°
 - XV. In 1st angle projection, top view always - (a) above the front view, (b) below the front view, (c) left side of the front view, (d) right side of the front view.

GROUP-B

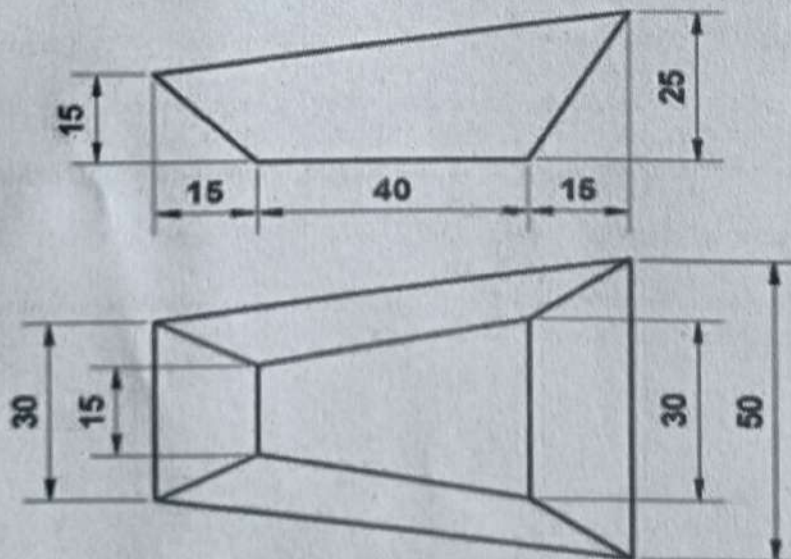
(Answer any Five (05) questions.) (8 x 5 = 40 marks)

2. Draw a full sectional front and top view of following diagram.

(4+4 = 8)

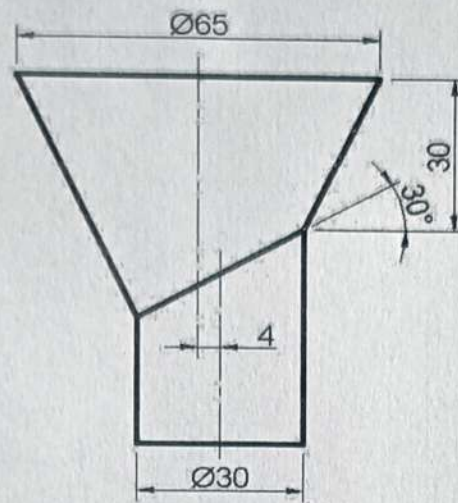


3. An oblique pentagonal prism, having base 30 mm side and height 70 mm long axis, has the axis inclined at 60° to the base and is resting on its base on HP such that one of the rectangular faces is parallel to the VP. Draw the development of its lateral surfaces. (8)
4. A vertical cone, base diameter 75 mm and axis 100 mm long, is completely penetrated by a cylinder of 45 mm diameter. The axis of the cylinder is parallel to HP and VP and intersect axis of cone at a point of 25 mm above from base. Draw projections showing curves of intersection. (8)
5. The Plan and Elevation of a tray are as shown in the figure below. Develop the surface of the tray. (8)



6. Develop the lateral surfaces of these transition pieces.

(4+4= 8)



7. Draw only the sectional elevation of a socket & spigot joint considering the following dimensions:

(8)

- Dimensions of both rods = 25 mm
- Diameter of collar of spigot = 50 mm
- Thickness of collar of spigot = 15 mm
- Outside diameter of spigot = 32 mm
- Length of spigot after the collar = 85 mm
- Diameter of collar of socket = 65 mm
- Thickness of collar of socket = 25 mm
- Inside diameter of socket = 32 mm
- Length of cotter = 100 mm
- Thickness of cotter = 6 mm
- Taper of cotter = 1 in 30

8. Draw full sectional front view of a protected type rigid flange coupling having diameter of the shaft diameter $D=40$ mm. Take other dimensions from the standard proportion are also given below

(8)

- Length of hub = $1.5D$
- Outside hub diameter = $2D$
- Flange diameter = $4D$
- Flange width = $0.8D$
- Flange thickness = $0.5D$
- Thickness of the protective circumferential flange = $0.25D$
- Number of bolts = 4
- Pitch circle diameter of the bolts = $2.8D$

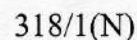
Missing dimension may be assumed (if any).

9. Draw neatly and proportionately a spur gear of 24 teeth, P.C.D. = 300 mm. pressure angle = 20° , showing at least three number of teeth profile.

(8)

10. a) Define tolerance. What is mean by Hole basis and Shaft basis system. b) write short notes (any two)
i) clearance fit ii) interference fit iii) transition fit.

$[(1+3)+(2+2)]$



3200682

Time Allowed: 3 Hours

Full Marks: 60

Answer any Five (05) Questions from Group-B.

GROUP-A

-



3200682

- (xii) For development of lateral surface of a open-ended right-circular cylinder of radius 50 mm and height 80 mm, the size of sheet-metal requirement (in mm^2) will be (a) 2π (b) π (c) 0.2π (d) 0.02π .
- (xiii) Detailed drawing of each part of a machine is called (a) part drawing (b) assembly drawing (c) patent drawing (d) tabular drawing
- (xiv) When the line of intersection formed by two solids is curve, then the solids are (a) pyramid vs. cuboids (b) prism vs. cube (c) pyramid vs. cube (d) Cone vs. Cylinder
- (xv) Bill of Materials is a part of (a) Part drawing (b) assembly drawing (c) blue-print (d) nowhere.

GROUP-B

All drawings should be done following first-angle projection method only.

2. A square prism of side of base 40 mm and axis 70 mm is resting on its base on H.P., such that one rectangular face of it is parallel to V.P. It is cut by a section-plane, perpendicular to V.P. and inclined at 45° to H.P. The section-plane is passing through the top of an edge perpendicular to V.P. Draw the development of the lateral surfaces of the truncated prism, assuming suitable scale. [8]
3. An air-conditioning duct connects a circular section of 40 mm diameter with a square section of 70 mm side through a 75 mm long transition piece. Draw the development of the lateral surface of the transition piece. Both sections of the transition pieces are coaxial and it is resting on its square base on the HP. [8]



4. A vertical cylinder of diameter 70 mm is completely penetrated by another horizontal cylinder of diameter 50 mm, their axes bisecting each other at right angles. Draw their projections showing curves of intersection, assuming the axis of the penetrating cylinder is parallel to parallel to both H.P. and V.P. Height/ length of the cylinders may be assumed suitably. [8]

5. According to the relative position of the tolerance zones of coupled parts, classify fit. In a limit system, the following limits are specified for the mating parts.

- 0.005

+ 0.020

Shaft: $30^{-0.018}$ mm

Hole: $30^{-0.000}$ mm

Determine (a) basic size (b) shaft and hole tolerances (c) the shaft and hole limits (d) type of fit.

[2 + (1+2+2+1) = 8]



An isometric drawing of a mechanical component, likely a bracket or a part of a machine. The part has a base plate with a central slot and two side flanges. The dimensions are as follows:

- Overall width: 60
- Overall height: 12
- Central slot width: 40
- Slot depth: 12
- Side flange width: 12
- Side flange height: 12
- Side flange thickness: 12
- Side flange hole diameter: 16
- Side flange hole offset from edge: 12
- Side flange hole offset from center: 16
- Side flange hole offset from bottom: 20
- Side flange hole offset from top: 20
- Side flange hole offset from right: 20
- Side flange hole offset from left: 20
- Side flange hole offset from front: 20
- Side flange hole offset from back: 20

A coordinate system is shown with the X-axis pointing to the right.

[8]

[NEATNESS IS HIGHLY RECOMMENDED]

MECHANICAL ENGINEERING DRAWING*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 is compulsory, this answer is to be made in separate loose answer script(s) provided for the purpose. The loose sheet must be stapled with drawing sheet at the end of examination. In Question No. 1 (Group – A), out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for explanation of the answer.

&

Answer any Four (04) Questions from the rest (Group – B).

GROUP-A

Answer any Ten (10) questions from Question No. 1

1. Choose the correct answer from the given alternatives and explain your answer (any ten): **2 x 10 = 20**
 - i) Knuckle joint is more rigid than socket and spigot joint. (a) True or (b) False
 - ii) Which development method is used for development of cone ? - (a) Parallel line method, (b) Approximation method, (c) Radial line method, (d) none of these
 - iii) When a plane is parallel to a reference plane, its projection on that plane is – (a) a line, (b) a plane, (c) a solid, (d) none of these.
 - iv) When a clockwise rotation of a nut moves towards the bolt head, the thread is said to be left hand thread. (a) True or (b) False
 - v) When two intersecting cylinders are of same diameters, then the lines of intersection will be – (a) two straight line, (b) one straight line, (c) one circle (d) none of these.
 - vi) In spur gears – (a) both shafts are parallel, (b) teeth are straight, (c) teeth are parallel to axis, (d) all of these.
 - vii) In spur gear, addendum is the radial height of a tooth above the – (a) addendum circle, (b) dedendum circle, (c) pitch circle, (d) none of these.
 - viii) In hole basis system – (a) lower deviation of the hole will be zero, (b) upper deviation of the hole will be zero, (c) both (a) & (b), (d) none of these.
 - ix) Crowning of a pulley is done to – (a) increase strength of the pulley, (b) avoid slipping of the belt, (c) prevent the wear and tear of the belt, (d) all of these.
 - x) The device used to connect two shafts together at their ends for transmission of power is known as – (a) gear, (b) coupling, (c) pulley, (d) all of these.
 - xi) The joint commonly used to convert rotary motion into reciprocating motion or vice versa is – (a) gib & cotter joint, (b) knuckle joint, (c) socket & spigot joint, (d) sleeve & cotter joint.
 - xii) Which one is the female part of socket & spigot joint? – (a) spigot, (b) socket, (c) cotter, (d) none of these.
 - xiii) An intentional difference between the hole dimension and shaft dimension for any type of fit is called – (a) limits, (b) allowance, (c) tolerance, (d) none of these
 - xiv) In the hole and shaft pair size $\Phi 50H7/c8$, the number 7 & 8 indicate – (a) tolerance grade, (b) accuracy, (c) surface texture, (d) none of these.

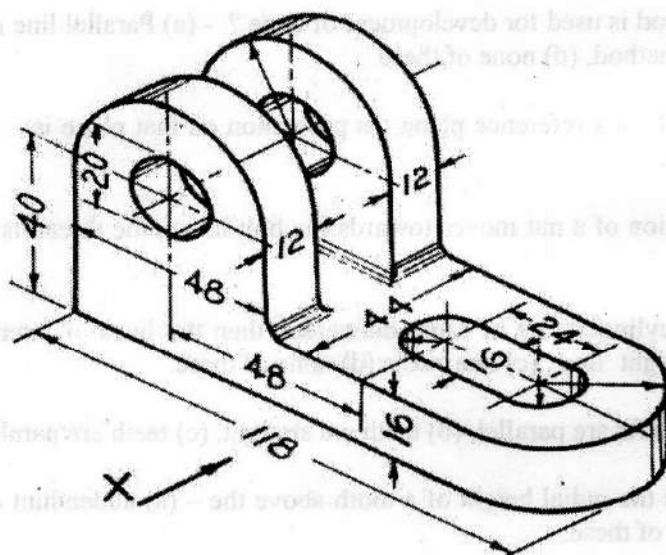
- xv) The development sketch for the lateral surface of a cone involves – (a) an arc of length equal to the radius of the base, (b) an arc of length equal to the circumference of the base, (c) a cord of length equal to the circumference of the base, (d) an arc of length equal to the true length of the slant generator.

GROUP-B

Answer any Four (04) questions.

4 x 10 = 40

2. Draw the development of an oblique cone of diameter 50 mm and axis 70 mm long, having the axis inclined at 60° to the base is resting on HP on its base with its axis parallel to V.P. 10
3. A vertical cylinder of 50 mm diameter and height 75 mm is resting on its base on horizontal plane is completely penetrate by another similar cylinder (same diameter and length). Their axes bisect each other at right angle and parallel to V.P . Draw their projection showing lines of interpenetration on the two cylinders. 10
4. Draw front view from 'X' and top view (with dimension) of the object of which isometric view is given below. All dimensions are in mm. 10

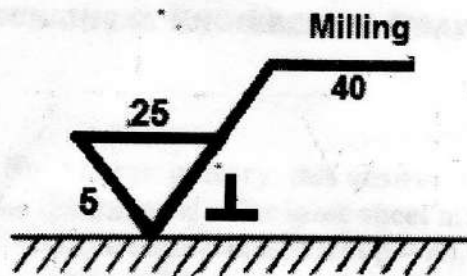


5. Draw neatly and proportionately a spur gear of 30 teeth , P.C.D = 300 mm , Pressure angle = 20° , showing at least three numbers teeth profile and some important parameters such as pressure angle, addendum, dedendum, pitch circle etc. 10
6. Draw only front view (elevation) of a knuckle joint having the pin 25 mm diameter. Assume the dimensions of other parts of the knuckle joint as per proportions. 10
7. Draw a sectional front and top view of a socket & spigot joint where inner diameter of the pipe is 150 mm and Maximum outer diameter is 280mm. Assume suitable dimensions for other parameters. 10
8. Draw half-sectional front view and side view of a protected type flange coupling having shaft diameter (D) = 50 mm. Some of the standard proportions are also given below:
 - a. Outside hub diameter = 2D,
 - b. Length of the hub = 1.5D,
 - c. Flange diameter = 4D,
 - d. Width of the flange = 0.8D,
 - e. Thickness of the flange = 0.5D,
 - f. Number of bolts = 4
 Missing dimensions, if any, may be assumed.

10

9. a) State the meaning of the figure shown in below -

1 x 1



b) Draw the symbols for the following features which are controlled in geometrical tolerance :

1 x 1

- i) Flatness
- ii) Cylindricity.
- iii) Angularity
- iv) Parallelism
- v) Concentricity

2909

June 2022

318/1(N)

MECHANICAL ENGINEERING DRAWING*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. Lateral surface of a cone can be developed by -
i) parallel line development method, ii) conical method iii) radial line development method
iv) approximate method
- ii. Which of the followings is/are never shown in sections? -
(a) Rivets (b) keys (c) bolts (d) all of these
- iii. In conventional representation of Cast Iron, the angle of hatching line with the longitudinal axis is -
(A) 0 degree, (B) 35 degree, (C) 45 degree, (D) 90 degree.
- iv. Detailed drawing of each part of a machine is called -
(a) Assembly drawing, (b) part drawing, (c) patent drawing, (d) tabular drawing
- v. Tolerance is the difference between- (a) maximum and minimum size (b) maximum and basic size
(c) basic size and minimum size (d) none of these
- vi. In a steam engine, the valve rod is connected to an eccentric by means of a -
(a) knuckle joint, (b) universal joint (c) flange coupling (d) cotter joint.
- vii. Following method is used for Development of Right Cylinder - (A) Radial line method (B) Parallel line method (C) Approximate Method (D) All of the above
- viii. When line of intersection formed by two solids is curve then the solids are-
(a) Cone, cylinder (b) Cube, Prism (c) pyramid, cube (d) pyramid, cuboids.
- ix. Sectional views reveal-
a) inner details, b) external features, c) overall size of the object, d) none of these
- x. The value of the ratio of isometric length to true length is -
a) 0.141, b) 0.372, c) 0.815, d) 0.642.
- xi. Blue prints are made for - a) working drawing assembly, b) designed assembly, c) installation assembly d) sub-assembly drawing
- xii. When the cutting plane cuts the entire object the section is known as-
(a) full section (b) half section (c) revolved section (d) removed section.
- xiii. Which type of solid has two bases of parallel and equal polygon?
(A) Pyramid, (B) Cone, (C) Cylinder, (D) Prism
- xiv. Basic Hole means-
(a) Upper deviation is zero, (b) lower deviation is zero, (c) both (a) & (b), (d) none of these
- xv. Standard tapers of a cotter used in industrial practice is -
a) 1 in 50, b) 1 in 30, c) 1 in 100, d) all of the above

2. Fill in the blanks (any ten):

1x10

- a) Base of an oblique cone is a _____
- b) Bill of materials is a part of _____ drawing.
- c) The full form of BIS _____
- d) LMC stands for _____.
- e) Most common material for pulley is _____.
- f) The development of a right prism will be _____.
- g) Hidden lines are used to show _____ edges of the object.
- h) The development of lateral surface of a pyramid consists of a number of equal _____ triangle in contact.
- i) To show the surface of section, hatching lines are drawing at _____°.
- j) Spigot is a part of _____ joint.
- k) Angle between the isometric axes is _____.
- l) Dedendum is the radial depth of a tooth below the _____ circle.
- m) Curves of penetration are found when _____.
- n) Symbol used to represent single sided fillet weld is _____.
- o) Intersection of one plane surface and a curved surface is always a _____ line.

3. Answer the following questions (any ten):

1x10

- a) Show the symbol of CI section in Engineering Drawing.
- b) H7 g6 – what type of fit is this?
- c) State the ratio of length to end width of an arrow.
- d) Define cutting plane of a sectional view.
- e) Distinguish between cotter joint and knuckle joint.
- f) Write the name of the most common limit gage used for inspecting the hole diameter.
- g) What do you mean by 'allowance' in limits and fits?
- h) What is a key?
- i) Socket is used in which joint?
- j) Name the three parts of a pulley.
- k) Show the V-weld symbol.
- l) Sugarcane Juice maker is an example of which type of fit?
- m) Which type of coordinate system is used in AutoCAD?
- n) Show the symbol of ply wood section in Engineering Drawing.
- o) Name three types of rigid shaft couplings.

Group B

4. Answer the following questions (any six):

2x6

- i. Define taper with example.
- ii. What is the principle of development of surfaces?
- iii. Draw a sketch of interference fit.
- iv. Which method of development is employed in case of cones?
- v. In '50H7/c8' - what is represented by the number 50?
- vi. Draw (a) centre lines (b) visible lines (c) hidden lines.
- vii. What is an oblique cylinder?
- viii. Why the face of a pulley is crowned?
- ix. What are the basic requirements of a part drawing?
- x. What are the three classes of fits as per BIS system?

Group C

5. Answer the following question (any one):

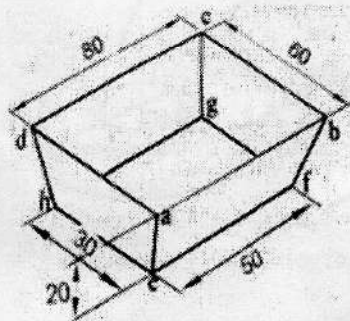
1x6

a) A vertical square prism, base 40 mm side, is completely penetrated by a horizontal square prism, base 25 mm side, so that their axes intersect. The axis of the horizontal prism is parallel to the VP. while the faces of the two prisms are equally inclined to the VP. Draw the projections of the solids, showing lines of intersection. Taking lengths for the prisms as 50mm.

6

b) Draw the development of the surfaces of the given tray.

6



c) Give a neat sketch of two cylinders being penetrated with each other at their axes at right angle.

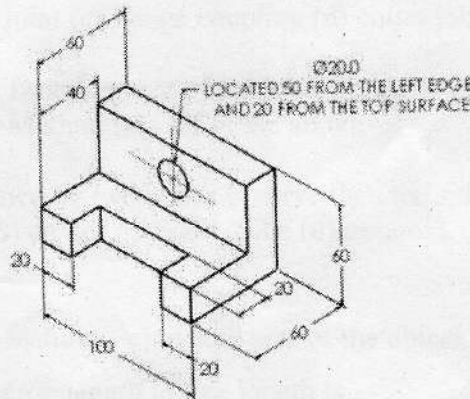
6

6. Answer the following question (any one):

1x6

a) Draw the orthographic views (front and top) from the Isometric view.

6



b) What is meant by hole basis and shaft basis system? Which one is preferred and why?

3+3

c) What is Thread terminology? Explain it with a suitable figure.

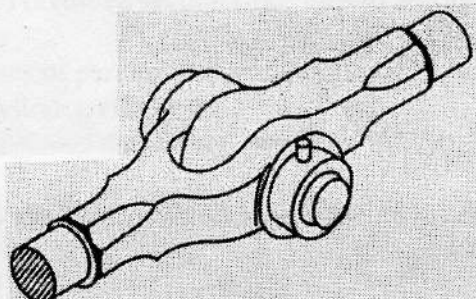
2+4

7. Answer the following question (any one):

1x6

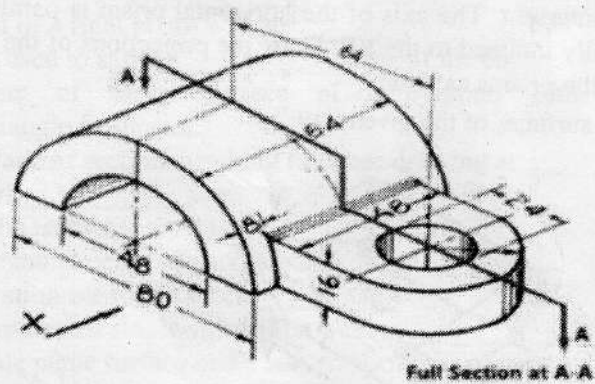
a) Make neat freehand sketch orthographic view of each part of the following assembly of a knuckle joint (only elevation) and name them.

6



b) Draw the profile of involute teeth for a spur gear having 25 teeth and module of 10 mm, and assuming pressure angle of 20° . (Draw at least 3 teeth) 6

c) Draw the Full sectional elevation (i.e., section at A-A) viewed from X and right-hand side view as shown in figure. 6



March 2021

MECHANICAL ENGINEERING DRAWING

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and Answer any three questions from the rest.

1. Answer the following questions (any eleven):

11x1

State whether the following statements are True or False:

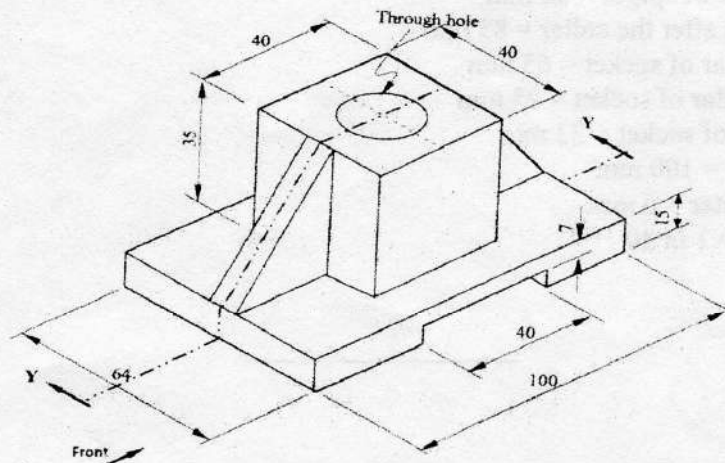
- i) A knuckle joint is used for rods which are in a straight line.
- ii) BSW thread means Bureau of Standard Whitworth thread.
- iii) The 'v' threads have more friction and are stronger than the square threads.
- iv) Grub screw is used to prevent the rotation of brasses in a bearing.
- v) Screw jack has square thread.
- vi) The big end of a connecting rod is the end fitted to the gudgeon pin.
- vii) Involute of a circle is used as the profile of a gear tooth.

Fill in the blanks:

- viii) _____ joint is used for rods in tension or compression.
- ix) _____ is the radial height of a tooth above the pitch circle.
- x) The number of units of PCD per teeth is called _____.
- xi) In case of saddle key _____ keyway is provided on the shaft.
- xii) A machine spindle can be driven at different speeds by fixing _____ on the spindle.
- xiii) Foundation bolts are used for _____ machines to their foundations.
- xiv) _____ bearing is used to supports for the vertical rotating shaft.
- xv) All lines and curves in the development represent only the _____ dimensions.

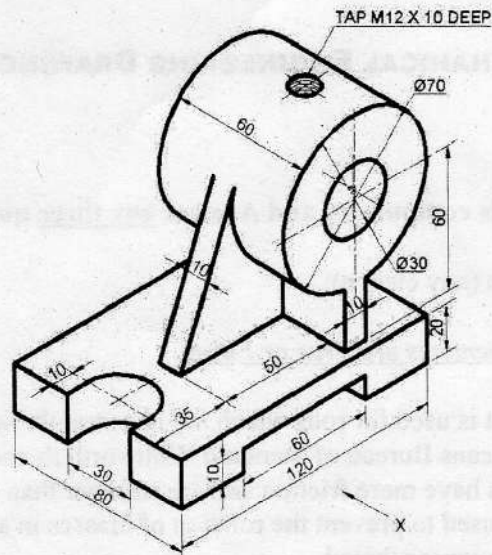
2. Draw a sectional front, top and left hand view of following diagram.

8

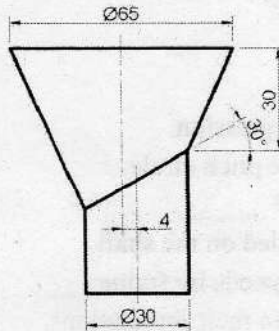


- 3. A vertical square prism, base 40 mm side, is completely penetrated by a horizontal square prism, base 25 mm side, so that their axes intersect. The axis of the horizontal prism is parallel to the V.P. While the faces of the two prisms are equally inclined to the V.P. draw the projections of solids, showing lines of intersection. Assume suitable lengths for the prisms. 8
- 4. Draw neatly and proportionately a spur gear of 20 teeth, P.C.D. = 300 mm. pressure angle = 20° , showing at least three nos. of teeth profile and some important parameters such as pressure angle, addendum, dedendum, pitch circle, face, flank, etc. 8

Draw full sectional elevation & plan of the following isometric view. The mark 'X' indicates front side. 8



6. Develop the lateral surfaces of these transition pieces. 8



7. Draw only the sectional elevation of a socket & spigot joint considering the following dimensions: 8

- Dimensions of both rods = 25 mm
- Diameter of collar of spigot = 50 mm
- Thickness of collar of spigot = 15 mm
- Outside diameter of spigot = 32 mm
- Length of spigot after the collar = 85 mm
- Diameter of collar of socket = 65 mm
- Thickness of collar of socket = 25 mm
- Inside diameter of socket = 32 mm
- Length of cotter = 100 mm
- Thickness of cotter = 6 mm
- Taper of cotter = 1 in 30

December 2019

MECHANICAL ENGINEERING DRAWING

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.

Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.

Answer any three questions from the rest.

1. A. Fill in the blanks (attempt all):

1x6

- i) A screw is specified by _____ diameter.
- ii) In gear circular pitch $\times$ No. of teeth = _____ $\times \pi$.
- iii) Pitch circle is _____ than base circle.
- iv) Bench vice has _____ thread.
- v) A small size screw external thread is often cut by means of a tool called _____.
- vi) Addendum circle is _____ than dedendum circle.

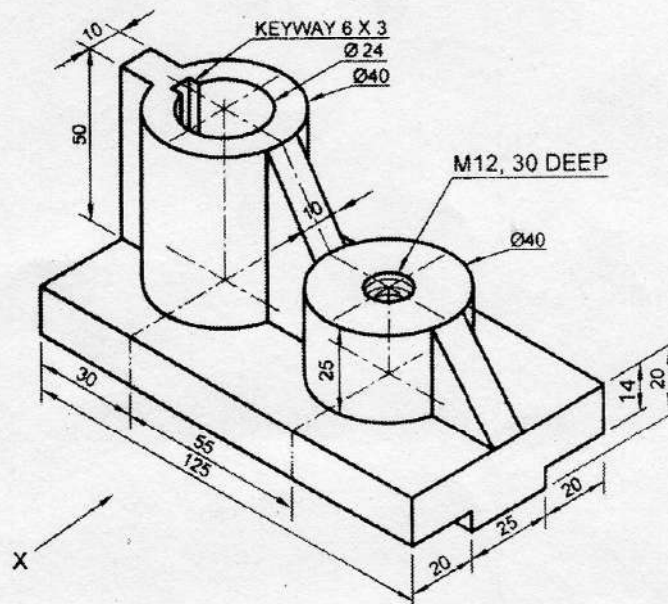
B. State whether following statements are True or False:

1x5

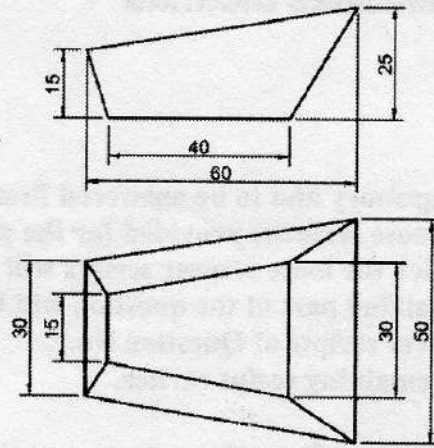
- i) Section lines are drawn approx. 2 mm gaps.
- ii) Module is the ratio of No. of teeth to PCD.
- iii) Pressure angle of involutes spur gear is usually taken as 20° .
- iv) The ratio of isometric length and true length is $\sqrt{2} : \sqrt{3}$
- v) Pressure angle is an angle between line of action and common tangent to the pitch circle.

2. A right cylinder of base $\phi 70$ mm and axis 100 mm long is completely penetrated by another cylinder of $\phi 40$ mm. the axis of the smaller cylinder is parallel to both the HP & VP and intersects at the midpoint of the bigger cylinder. Draw projections showing curves of intersection. 8

3. Draw only full sectional elevation of the following isometric view. The mark 'X' indicates front side. 8



4. A single stainless-steel sheet is to be cut into some pieces to fabricate a sheet metal tray. Develop the lateral surfaces of the following figure of the tray. 8



5. Draw neatly and proportionately a spur gear of 25 teeth, P.C.D. = 600 mm. pressure angle = 20° , showing at least three nos. of teeth profile and some important parameters such as pressure angle, addendum, dedendum, pitch circle, face, flank, etc. 8

6. Draw elevation, plan and side view of a hexagonal headed bolt & nut with a washer adopting the following dimensions. 8

- Diameter of bolt (d) = 20 mm
- Length of shank = 100 mm
- Thickness of head of the bolt = $0.8d$
- Length of the threaded part of the bolt = 75 mm
- thickness of the nut = d
- Width across flats of hexagonal bolt and nut = $1.5d + 3$ mm
- Width across corners of hexagonal bolt and nut = $2d$
- Angle of chamfer = 30°
- Radius of chamfer arc = $1.4d$
- Diameter of washer = $2d + 3$ mm,
- Thickness of washer = $0.12d$

December 2018

MECHANICAL ENGINEERING DRAWING

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.

Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.

Answer any three questions from the rest.

One mark is reserved for Draftsmanship.

1. Answer the following questions (any ten):

1×10

Fill in the blanks:

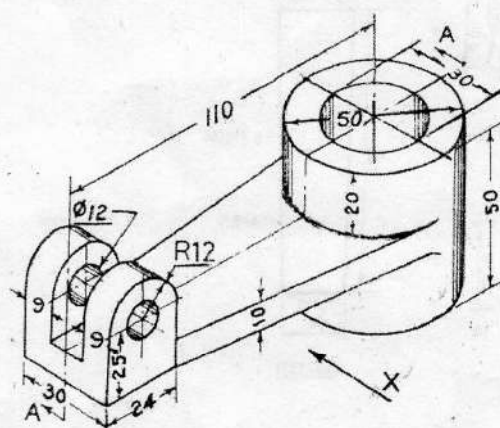
- i) A _____ coupling permits within certain limits, relative rotation and variation in the alignment of shafts.
- ii) The three parts of a pulley are hub, arms and _____.
- iii) _____ is the number of teeth per unit length of P.C.D.
- iv) The curve along which the two lateral surfaces meet each other is known as _____.
- v) Sphere and ellipsoid are the _____ curved surface solids.
- vi) When axis of any solid is inclined to its base, the solid is called _____ solid.
- vii) _____ method is applied to develop the lateral surfaces of transition pieces and oblique pyramid.

State whether the following statements are True or False:

- viii) Buttress thread is a combination of triangular and square threads.
- ix) A cotter is used to connect rigidly two rods subjected to shear.
- x) A knuckle joint is used for joining D-slide valve and eccentric rod of a steam engine.
- xi) In the fast and loose pulley, the fast pulley is free to rotate on the shaft.
- xii) In journal bearing the bearing pressure is parallel to the axis of the shaft.
- xiii) Bushes are prevented from rotation by using snug at the bottom.
- xiv) Engine cross head acts as a connecting link between the piston rod and the connecting rod.

2. Draw sectional front view and top view of the following Fig.1.

8

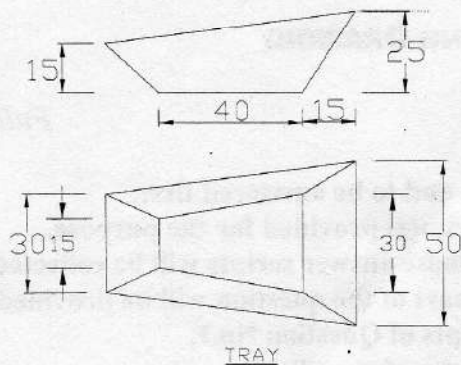


Front

Fig.1

3. Draw the lateral surface development of a tray shown in Fig.2.

8



4. A vertical cylinder of 70 mm diameter and 120 mm long resting on HP with one of its bases. It is completely penetrated by a square prism of 40 mm side of base and 120 mm long. The axes bisect each other at right angle. Draw the curves of intersection when all the base sides of prism are equally inclined to HP. 8
5. Draw the profile of an involute teeth for a spur gear having 30 teeth and module equal to 10 mm. Assuming a pressure angle of 20° . 8
6. Draw full sectional front view and side view of a C.I. pulley with the following dimensions. 8
- Bore dia = 22mm
 - Hub dia = 45 mm
 - Length of hub = 36 mm
 - Dia of pulley = 160mm
 - Width of pulley rim = 40mm
 - Thickness of pulley rim = 5mm
 - Number of arms = 4
 - Taper arm at boss = 20mm $\times$ 12mm and at rim = 12 mm $\times$ 6 mm
 - Crowning = 1.5mm
 - Key 6 mm $\times$ 4 mm
- Missing data, if any may be assumed.
7. Draw the full sectional front view and side view of a protected type rigid shaft coupling having diameter of shaft is 100 mm. Take standard proportions for other dimensions. 8
8. Fig-3 shows the details of spigot and socket joint. Assemble the parts and draw the sectional front view and side view. 8

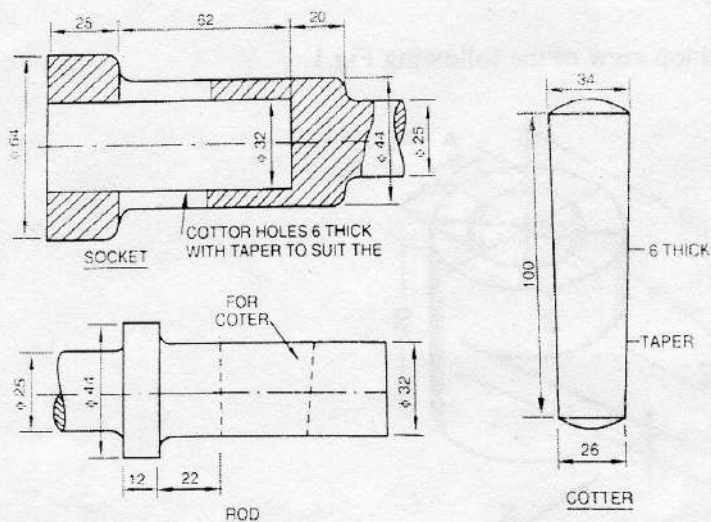


Fig-3